

MyPiggyBank

Project Type C – User-Driven Software Engineering Course

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Contents

1	Introduction	3
2	Application Scenario	4
2.1	Why we created MyPiggyBank	4
2.2	Our Constraints (Project Requirements)	4
3	Work Plan	6
3.1	How we worked	6
3.2	How we collected feedback	7
4	Requirements Analysis	8
4.1	User Requirements	8
4.1.1	User Profile	8
4.1.2	Personas	9
4.1.3	Persona 3: Ali	13
4.1.4	Scenarios	17
4.2	User Analysis	19
4.2.1	Questionnaire Design	19
4.2.2	Sample Description	21
4.2.3	Results	22
4.2.4	Key Takeaways	29
4.3	Competitor Analysis	30
4.3.1	MoneyBox	30
4.3.2	Revolut	30
4.3.3	Physical Piggy Bank	30
4.3.4	Comparative Overview	30
4.4	Hierarchical Task Analysis (AS-IS)	31
4.5	Conclusions of the Analysis	34
5	Design and Implementation	36
5.1	Top-Level Navigation STN	36
5.2	Sub-STN 1: Expense Entry	37
5.3	Sub-STN 2: Piggy Bank Management	38
5.4	Sub-STN 3: Reports and Summaries	39
5.5	Sub-STN 4: Settings	40
5.6	Sub-STN 5: Spending Alert Flow	41
5.7	Mapping to Functional Requirements	42
5.8	From HTA to STN: The TO-BE Perspective	43

6	Prototypes	44
6.1	Prototype Zero	45
6.1.1	Mockups and Interaction Flows	45
6.1.2	Expert Evaluation	48
6.1.3	From Prototype Zero to Prototype One	51
6.2	Prototype One	51
6.2.1	Corrections Implemented	51
6.2.2	Qualitative Evaluation: Think-Aloud Study	55
6.2.3	Controlled Experiment	55
7	Final Realization	58
7.1	Features	58
7.2	Strengths	58
7.3	Weaknesses	59
7.4	Improvements Over the Previous Version	59
8	Conclusions and Future Work	60
8.1	Key Points of the Realization	60
8.2	Critical Phases and How They Were Overcome	60
8.3	Reflection on the HCI Techniques Used	61
8.4	Future Work	61
9	Bibliography	62
10	Appendix	63
10.1	Heuristic Evaluation — Full Form	63
10.2	Cognitive Walkthrough — Full Form	64
10.3	Think-Aloud — Task List and Raw Observations	65
10.4	Controlled Experiment — Raw Data and Analysis	66

Chapter 1

Introduction

MyPiggyBank, even before being an app, is an innovative concept designed for those who do not want to waste their money—it is not just a typical financial application. This app is meant to serve as a true companion for those who genuinely want to start saving and achieve their goals and desires.

We envisioned it as a place where you can create digital piggy banks, log expenses manually (a crucial part, as financial awareness stems from there), monitor progress toward set targets with clear charts, and receive alerts if you are overspending on extras. The key to everything is speed: it takes only a few seconds—just the time of a coffee break—to integrate it perfectly into your day.

In this project, we wanted to combine utility (features like multi-currency management and progress tracking) with the psychological side of saving (the motivation and mindset behind every penny saved). By merging these two aspects, we believe we can reach both the hearts and minds of our users. Another fundamental component is the design, where every screen and interaction has been crafted to make the experience as fluid and enjoyable as possible.

In the following pages, we will explore how we moved from idea to reality, following a path that puts the user at the center of everything: from the initial scenario study to testing with real people and the final version.

Chapter 2

Application Scenario

2.1 Why we created MyPiggyBank

Today, spending money has become an almost invisible gesture: a tap on a phone, a click on a computer, and it's done. The problem is that when you don't physically handle cash or have little oversight of your outflows, you lose the sense of what you are actually paying. It becomes difficult to understand how that small expense today can concretely affect your dreams for tomorrow. Saving, ultimately, is not just a matter of numbers, but of mindset, consistency, and habits.

We focused primarily on those who—like students with limited funds, freelancers, or young professionals—find themselves managing fluctuating incomes and a thousand different goals (the dream trip, an unforeseen emergency, that special gift). For them, a rigid plan doesn't work: they need something that adapts to changing daily priorities, something flexible designed specifically for the individual user.

In this market, alternatives like banking apps or classic Excel sheets already exist, but something was missing that combined three fundamental ingredients:

1. Manual entry that doesn't waste time.
2. Graphics that provide instant clarity on your financial direction and behavior.
3. Total respect for privacy without the requirement of linking a bank account.

This is where **MyPiggyBank** comes in. We don't want to be a bank, nor do we want to automate everything: we want the user to be in the driver's seat. By recording every expense mindfully, we want to help people rediscover the value of the old-fashioned ceramic piggy bank, but in a modern, interactive, and significantly more stimulating digital form.

2.2 Our Constraints (Project Requirements)

- **Speed above all:** If logging an expense becomes a chore, you will eventually stop doing it. That's why every operation must be "lightning-fast": under 30 seconds and you're done.
- **Privacy is sacred:** We know how sensitive financial data is. This is why we

chose a *local-first* approach: user data stays on their device. Consequently, the app works perfectly offline, and no one but you can access your business.

- **Intelligent tips with real data:** Even without connecting to a bank, the app uses localized heuristics to learn from your habits. For example, the system calculates a "daily burn rate" (the remaining category budget divided by the remaining days in the month) to trigger non-intrusive alerts when a user's spending velocity threatens a savings goal.
- **No borders (Multi-currency):** For those who travel or work with international clients, managing different currencies should be effortless. To maintain our strict local-first constraint without external server pings during transactions, the system caches the latest exchange rates locally whenever the device connects to Wi-Fi. Conversions are timestamped for transparency, ensuring full offline capability without sacrificing accuracy.
- **Accessibility:** For now, we have concentrated on a highly visual and interactive interface. We recognize this is a starting point and would like to make it even more accessible to everyone in the future.

Chapter 3

Work Plan

3.1 How we worked

We divided the work into three major stages:

1. **Problem Identification:** In the first phase, we identified the core issues and the general context. We considered the macro-profiles of target users, their money management habits, and the difficulties they encounter when saving. This step was necessary to clarify the project's direction and the main constraints.
2. **Requirements analysis:** This part involved defining personas and usage scenarios, analyzing competitors, and studying user needs through tools like questionnaires. This allowed us to understand potential customers' needs; for example, surveys showed that people wanted weekly goals in addition to monthly ones.
3. **MyPiggyBank Optimization:** Once the data was gathered, work continued with the design of the application's structure and the refinement of previous ideas, integrating new concepts by addressing potential user requests.

Phase		What we did
Context	Analysis	We dug into the problem, identified our audience, and set the project boundaries.
Requirements Gathering		We listened to people through questionnaires, created user personas, and analyzed competitors.
Design		We mapped out the app and decided how the user should navigate between functions.
Prototype		Initial drawings (wireframes) to give concrete form to the structure.
Final Version		Final design polishes and the drafting of this document.

Table 3.1: MyPiggyBank Project Planning

3.2 How we collected feedback

Online Questionnaire

Through a survey with multiple questions, over 30 people shared how they save. The questionnaire was designed to uncover issues related to saving, using an interactive nature that generated new questions based on previous answers.

Chapter 4

Requirements Analysis

4.1 User Requirements

4.1.1 User Profile

In this section, we define the core characteristics and behavioral ranges of the primary user group for MyPiggyBank. This profile helps to identify the boundaries of the system’s design. The values reported here represent the *intended target*; the actual sample collected through the questionnaire is described and compared with this profile in Section 4.2.

Age:	18–45 (Average: 25 years)
Job Titles:	Primarily Freelancers, Students, Junior Professionals, Remote Workers, and Young Full-time Employees
Experience Level:	0–15 years in the workforce (typically 3–4 years)
Work Hours:	20–50 hours/week; often includes irregular or flexible schedules
Education:	High School diploma to PhD (Typical: some college)
Location:	Global (predominantly urban/suburban area with smartphone access)
Technology:	Daily smartphone use; familiar with mobile banking; relies on apps for lifestyle
Disabilities:	No specific limitations
Family:	Single, Roommates or Young Couples
App Usage:	2–3 times daily (usually after a purchase or at end-of-day); prefers 30-sec interactions
Privacy Profile:	Prefers local data storage or encrypted manual logs
Motivation:	Driven by clarity and overview of personal spending

Table 4.1: Primary User Characteristic Ranges

4.1.2 Personas

Persona 1: Chiara

Backstory: Chiara graduated in Design three years ago and has been working as a freelance graphic designer for two. She has learned to manage the technical side of the profession (quotes, invoices, tax deadlines) but her relationship with money remains primarily emotional: when a client pays late, she gets anxious; when a major wire transfer arrives, she treats herself to a dinner out to celebrate. She lives alone in a studio apartment in Milan, and a significant portion of her expenses is social: aperitifs with her university group, dinners with coworkers from the coworking space, weekend getaways. She does not consider these expenses "superfluous"; she sees them as part of her work (networking) and her identity.

Mindset: Chiara does not want to be judged by the app. Two years ago, she tried a banking solution that sent her notifications like "you spent 30% more on restaurants this week" and uninstalled it after ten days because it made her feel guilty. She wants a tool that shows her *what* is happening, not one that tells her *what* to do. She thinks in terms of concrete goals (the MacBook, the trip to Japan next year), but keeps them mentally separate from daily expenses, and this is exactly where things get out of hand: at the end of the month, she is surprised by how little she has saved, even though every single expense felt justified at the moment.

Context of use: Morning in the coworking space, a quick lunch out, afternoon in a client's studio, evening out with friends. On a typical day, she opens her phone dozens of times, but almost always for work notifications. A savings app for her must fit into the interludes: 20 seconds waiting in line for coffee, 30 seconds on the tram. Anything that requires sitting down and "doing the math" is postponed and then forgotten.

Persona Name:	Chiara
Quote:	“I want my savings to grow as fast as my portfolio, without the headache of spreadsheets.”
Age:	26 years old
Gender:	Female
Location:	Milan, Italy
Occupation:	Freelance Graphic Designer
Personality:	Self-confident, dutiful, socially active, highly organized, prefers quick mobile interactions that fit her busy routine
Tech Profile:	Expert smartphone user; relies on social media and professional apps for work
Motivations:	Wants to manage her savings more effectively, maintain financial stability despite irregular freelance income, and save for high-end hardware such as a new MacBook
Social Habits:	Frequently travels and dines out with a large circle of friends
Financial Needs:	Needs a flexible tool that can handle multiple currencies and support dynamic financial planning, with interactions short enough to fit between meetings
Frustrations:	Struggles to track small daily expenses that affect long-term goals; finds traditional banking apps too rigid for fluctuating monthly income; fears overspending during socially active periods

Table 4.2: Persona profile: Chiara

Persona 2: Marco

Backstory: Marco is 21 years old and is in his third year of a Bachelor's degree in Economics in Rome. To avoid weighing entirely on his parents' shoulders, he works weekends as a waiter in a busy trattoria in Trastevere. His income is a patchwork of a small monthly allowance from his family and cash tips that vary wildly depending on the weekend weather and the generosity of tourists. This cash component is his financial blind spot: because it doesn't pass through a bank account, it feels like "free money." He lives in a shared apartment with three other students near Piazza Bologna, and his expenses are deeply tied to student life: cheap university cafeteria meals mixed with sudden, unplanned craft beer rounds, streaming subscriptions shared with roommates, and train tickets to go back home on holidays.

Mindset: Marco feels a constant, low-level anxiety about money, but his coping mechanism is avoidance. He avoids checking his banking app because he dreads seeing a low balance. Like Chiara, he hates being lectured, but while Chiara rejects judgment out of a desire for independence, Marco rejects it because he already feels guilty about his lack of discipline. He has tried budgeting with an Excel sheet he copied from an online tutorial, but typing numbers into a cell on his laptop felt too much like studying, so he abandoned it after four days. He thinks in terms of immediate availability: if his card doesn't get declined at the supermarket checkout, he considers himself "fine." However, he has a burning desire to join his friends on a summer surf trip to Portugal, and he knows that without a structural change, he will be the only one left behind.

Context of use: Late morning in a crowded lecture hall, sprinting between university buildings, or during a chaotic 10-minute break between lunch and dinner shifts at the restaurant. Marco lives his life in fragments of chaotic activity. He opens his phone to check messages or social feeds during professor changes or while riding the Metro B. For an app to work for him, transaction logging or checking his allowance must happen in a single, frictionless gesture—ideally while balancing a backpack or a tray in his mind. If an action takes more than two taps, his attention shifts back to his notifications.

Persona Name:	Marco
Quote:	“I want to know exactly what I can spend today without checking my balance in fear, or dealing with boring spreadsheets.”
Age:	21 years old
Gender:	Male
Location:	Rome, Italy
Occupation:	University Student (Part-time Waiter)
Personality:	Curious, a bit disorganized, socially engaged, motivated by short-term rewards but easily distracted
Tech Profile:	Heavy smartphone user; native to messaging, gaming, and social media apps; zero experience with formal fintech infrastructure
Motivations:	Wants to stop running out of money mid-month, manage fluctuating cash tips, and securely save enough for a summer surf trip to Portugal with his friends
Social Habits:	Frequently hangs out with university peers, shares flat expenses, and splits casual dinner/drink bills using peer-to-peer payment methods
Financial Needs:	Needs a lightweight, highly visual indicator of daily disposable income that accommodates irregular cash inflows without complex manual accounting
Frustrations:	Consistently loses track of micro-expenses (coffees, casual street food, metro tickets); feels alienated by traditional banking apps; struggles to bridge cash tips with digital balances

Table 4.3: Persona profile: Marco

4.1.3 Persona 3: Ali

Backstory: Ali is 24 and has lived in Bologna for three years. He works as a delivery rider and tops up his income with the odd job here and there: a house move, a bit of handyman work, usually paid in cash at the end of the day. What he earns swings a lot from one week to the next, so planning ahead is more of a struggle than a habit. He shares a double room on the outskirts of the city and keeps his fixed costs as low as he can: rent, the discount supermarket, the phone credit he needs to work. Every month he also sets aside a share for his family, to help his younger sister with her studies. That one matters to him a great deal.

Mindset: For Ali, money is mostly about stability and a bit of peace of mind, not status or treating himself. What he'd really like is a small cushion for when things go wrong: if the bike breaks or he loses a few days of work, he doesn't want to be caught short. Mainstream banking apps don't really fit him, because they assume a fixed salary and a normal current account with direct debits, and he has neither. When the cash comes in, or when he tops up his prepaid card, he tends to spend it straight away on the essentials and on the money he sends home, so by the end of the month there's not much left. He's looking for something that helps him build up that little cushion without making him feel bad on the weeks when he can't put anything aside. His concrete, motivating goal is to buy an e-bike, so he can do more deliveries with less effort.

Context of use: Ali is on his phone constantly, but usually with his hands busy on the handlebars and one eye on the battery. Between deliveries he has only a few spare seconds, and by evening he's tired. For an app to be of any use to him, it has to be quick: log a payment in a single gesture while he waits to pick up an order, or in a couple of taps before he falls asleep. If the interface asks for too many steps, or throws complicated charts at him, he closes it and goes back to the method he trusts: a bit of mental arithmetic and a scrap of paper in his pocket.

Persona Name:	Ali
Quote:	“My income changes every day. At the end of a shift I just want to see at a glance how much I can put aside for my goals, without getting lost in complicated sums.”
Age:	24 years old
Gender:	Male
Location:	Bologna, Italy
Occupation:	Gig worker / Delivery rider
Personality:	Pragmatic, resilient, family-minded and very adaptable; copes well with uncertainty but values predictability when he can get it
Tech Profile:	Mobile-first; comfortable with navigation maps, translation tools, and peer-to-peer payment apps
Motivations:	Wants to build a small fund for the unexpected (e.g. bike repairs), manage to send money home each month, and over time buy an e-bike that makes the work easier
Social Habits:	Spends little socially; cooks at home, shares a room to save money, and now and then grabs a cheap coffee with colleagues from the hub
Financial Needs:	Needs manual cash tracking and support for irregular income; visual feedback that doesn’t pass judgement and copes with zero-earning days
Frustrations:	Income he can’t predict; no way to use the banks’ automatic saving features; fintech apps built around a fixed salary; a wish for more peace of mind about the unexpected

Table 4.4: Persona profile: Ali

Persona 4: Giulia

Backstory: Giulia is 30 and lives in Turin. After a degree in economics she landed a permanent job as a project manager at a mid-sized company: a fixed salary that lands on the 27th of every month, no surprises. She has lived with her partner for two years, and they split the rent and the household bills. In the couple, she's the one who keeps an eye on the money: she tracks everything in a Google Sheet, already has a few months' worth of emergency fund put away, and has set up an automatic transfer to her savings account at the start of each month. On paper, she has it all under control. Her two big goals are the deposit for a flat she wants to buy with her partner and one proper trip a year.

Mindset: Unlike the others, Giulia's problem isn't a shortage of money; if anything, it's the opposite. The stability has made her a little complacent, and every small pay rise has quietly turned into higher spending: one more subscription, dinners out more often, the nicer gym. She does save, but it all ends up in one undifferentiated pile, so the individual goals have no face. Six months in, she doesn't feel any closer to the flat, and that takes the wind out of her. Her bank's app shows the balance but gives no sense of where any of it is heading. There's a practical annoyance too: keeping shared goals in sync with her partner is clumsy—who put in what, how far along they are. She doesn't want an app that treats her like a beginner; she wants a serious tool that makes the goals tangible and lets her steer the surplus wherever she decides, in one move.

Context of use: Giulia's routine is steady and predictable, the exact opposite of Marco's or Ali's. She checks her phone calmly: over coffee in the morning, on her lunch break, on the sofa in the evening. She doesn't need two-second micro-interactions snatched from the chaos; she can spare a few minutes. But she's demanding: if the app feels shallow or toy-like, she'll drop it, because she expects something that matches the way she thinks. Typically she opens it on a Sunday evening to take stock of the week, and now and then during the month when something extra comes in—a refund, a bonus—and she wants to assign it to a goal before it dissolves into everyday spending.

Persona Name:	Giulia
Quote:	“I earn a good salary and I put money away every month, but my goals always seem just as far off. I want to see myself getting closer to something.”
Age:	30 years old
Gender:	Female
Location:	Turin, Italy
Occupation:	Project Manager (permanent contract)
Personality:	Organized, level-headed, forward-thinking, focused on the long term; disciplined but prone to a slow lifestyle creep
Tech Profile:	Expert user; comfortable with productivity tools, spreadsheets, and traditional banking apps
Motivations:	Wants to save the deposit for a flat with her partner, treat herself to one proper trip a year, keep her emergency fund topped up, and above all give her savings a concrete meaning
Social Habits:	A settled, planned social life: dinners with friends, weekends away; shares expenses and goals with her partner
Financial Needs:	Needs to run several goals in parallel with a visual way to allocate her surplus; wants to set up goals shared with her partner; needs a long-term view that makes progress visible
Frustrations:	Her savings pile up in one undifferentiated account and the goals stay abstract; a quiet, creeping rise in her spending; banking apps show the balance but don’t motivate; coordinating couple goals is awkward

Table 4.5: Persona profile: Giulia

4.1.4 Scenarios

The following usage scenarios are based on the personas defined in the previous section and show how MyPiggyBank can support daily money management in realistic contexts.

Scenario 1: Chiara and everyday expenses

Chiara does not see saving money as a sacrifice, but as an engine for professional freedom. For her, putting money aside means being able to make choices with greater peace of mind: accepting an interesting design project even if it is paid later, buying better tools for her work, or traveling without financial guilt. Her cognitive model is strongly goal-oriented; she views a new MacBook Pro not as a luxury consumer desire, but as a critical capital investment in her freelance studio.

However, her daily routine constantly erodes this long-term clarity. Moving rapidly from her home desk to a morning meeting at a client's studio, grabbing a quick lunch on the go, working the afternoon from a busy coworking space, and ending the night with an aperitif with peers, her expenses are fragmented. Each transaction feels entirely justified in isolation: *"It is just a quick lunch, it will not really affect my Japan fund."* The misalignment only surfaces at the end of the month, when she realizes these unmonitored micro-transactions have quietly bottlenecked her savings velocity.

Chiara completely rejects prescriptive financial tools that track percentages and trigger patronizing notifications. She demands absolute autonomy over her capital but requires ambient awareness. For MyPiggyBank, this means a frictionless, non-intrusive micro-interaction framework. Immediately after a transaction, she needs a 5-second visual validation to see if her current spending vector matches her milestone trajectories. If the application demands manual data entry or feels restrictive, she will abandon it; if it provides instant, objective clarity, it seamlessly integrates into her short transit interludes.

Scenario 2: Marco and the end-of-month check

Marco experiences money as a tool for short-term lifestyle balance rather than long-term wealth accumulation. He operates without a formal financial strategy; his primary objective is minimizing cash-flow anxiety while maintaining an active social life with his university peers and saving for a summer surf trip to Portugal. His relationship with money is highly cyclical and avoidant: receiving his part-time salary triggers a period of relaxed spending, whereas the final week of the month brings a spike in anxiety, forcing him to constantly check his mobile banking app in fear of a declined card.

This lack of control is exacerbated by his fragmented environment. Sprinting between Economics lectures, commuting via Metro B, and running weekend shifts at a crowded restaurant in Trastevere, Marco executes dozens of casual financial decisions on impulse. This issue is highly critical when managing his cash tips: because this money bypasses his digital bank account, it acts as invisible capital that vanishes into casual street food, coffees, and shared drink rounds. Each transaction feels insignificant: *"It is only a two-euro espresso,"** or *"I will cover the round tonight and make up for it next week."** Marco does not want a professional accounting suite, dense data tables, or intimidating charts that feel like an academic extension of his university courses. MyPiggyBank must function as an effortless visual pulse. For Marco, the app must drastically lower the psychological barrier to tracking both digital spending and physical cash tips. By transforming raw numbers into a clear, glanceable indicator of his daily disposable income, the system can

intercept his avoidant behavior, turning short-term progress into a gamified motivator before he reaches his breaking point at the end of the month.

Scenario 3: Ali and managing unpredictable cash flows

Ali sees saving as a small safety net rather than a luxury or a career move. Putting aside even fifty euros in a month means he can fix the tool he works with, without borrowing money or giving up days of deliveries. His thinking is short-term and tied to family duty: the money he sends home at the end of the month comes first, and everything else follows.

His day is made up of tiny amounts: four euros for one delivery, a five-euro cash tip, twenty euros for a small move in the afternoon. With the money coming in like that, it's hard for him to get a clear picture of what he's actually earning and spending. By the end of the day he's left with a few coins in his pocket and a shifting figure on the prepaid card, and no real overview.

Ali doesn't want an app that forces a rigid monthly budget on him, because no two of his months ever look the same. He's after something that makes room for that variability without making him feel guilty: on a good day he wants to log the small surplus in a single tap and send it, visually, towards the e-bike fund or the family. On a leaner week, the app shouldn't fire off stressful warning notifications. The interaction has to be immediate: a quick entry of the cash he's earned while he waits to collect an order at the pizzeria, without having to link bank accounts he doesn't have. Only if it proves flexible and free of red tape will the app become a practical alternative to the sums in his head and the scrap of paper in his pocket.

Scenario 4: Giulia and managing long-term goals

Giulia doesn't save to get by; she saves to build something. Putting money aside is already a settled habit—the weak spot is elsewhere: giving those savings a face and making them visible. The big goal, the flat, sits so far off that after months of dutiful transfers she feels no sense of progress, and the early enthusiasm fades. For her, MyPiggyBank has to turn one anonymous pile into several distinct, visual piggy banks: the flat, the trip, the emergency fund.

On a Sunday evening she opens the app, sees at a glance how much is left to go on the flat, and moves the month's surplus across with a single tap. When something extra comes in, she assigns it straight away instead of letting it evaporate into day-to-day spending. The feature that wins her over is shared goals with her partner, so the two of them look at the same picture without having to rebuild it on a spreadsheet every time.

The app should also help her see, honestly but without lecturing, how much of that surplus is slipping into spending rather than into the goals—her blind spot. For Giulia the value isn't chasing every cent; it's making real something that today is just a number parked in an account.

4.2 User Analysis

At this point we thought it appropriate to conduct a survey aimed at the intended user group in order to go beyond the application hypothesis alone but truly understand how much our idea was potentially appreciated or denigrated. The goal wasn't to validate a finished product, but to understand how potential users currently manage their money, what challenges they face when trying to save money, and whether the core ideas behind MyPiggyBank—manual expense entry, digital piggy banks, and spending alerts—actually match the real needs of the user group we targeted. As the main data collection method, we chose a web-based questionnaire (Google Forms). The survey was distributed via university group chats and personal contacts over the course of four days (April 3–6, 2026) and collected a total of **35 valid responses**.

4.2.1 Questionnaire Design

We organized the questionnaire into four macro-sections, each aimed at a different aspect of the user experience. In the first three sections, we decided not to mention our application at all, so as to avoid influencing the interviewees: we wanted to capture their real habits and frustrations before introducing the idea of a savings tool.

Below is the complete set of questions distributed to the participants, including the conditional logic applied to filter irrelevant items based on previous answers.

Section 1 — Demographics

Age group and income type, used to check whether respondents matched the user profile.

- **Q1. Which age group do you belong to?** (Single choice)
Less than 18 / 18–24 / 25–34 / 35–45 / More than 45
- **Q2. What is your main occupation?** (Single choice)
Student / Full-time employee / Freelancer / Part-time worker / Looking for work / Other
- **Q3. How would you describe your monthly income?** (Single choice)
Fixed and regular (e.g. monthly salary) / Fairly regular with some variation / Irregular or variable from month to month / I have no personal income at the moment

Section 2 — Current habits

How respondents currently track their spending, how often they check their spending, and how much control they feel they have over their finances.

- **Q4. Do you currently track your expenses in any way?** (Single choice)
No, I don't track them / I occasionally check my bank/card balance / I use my banking app's features (categorization, charts, etc.) / I use a dedicated budgeting app / I use a spreadsheet or notebook / Other
- **Q4b. (Conditional) Which app do you use?** (Open, optional)
Shown only if Q4 = "dedicated budgeting app".

- **Q5. How often do you check how much you have spent in the last week?** (Single choice)
Practically never / A few times a month / About once a week / Several times a week / Every day or nearly every day
- **Q6. How in control do you feel of your daily spending?** (Likert scale 1–5)
1 = Not in control at all, 5 = Completely in control
- **Q7. Has it ever happened that you reached the end of the month and realized you had spent more than expected?** (Single choice)
Yes, often / Sometimes / Rarely / Never
- **Q8. Which expenses do you find hardest to keep under control?** (Multiple choice, max 3)
Dining out, bars, aperitifs / Impulsive online purchases / Digital subscriptions / Transportation / Leisure activities / Daily groceries / I have no particular difficulty / Other

Section 3 — Savings and goals

whether respondents are saving for specific goals, what obstacles they encountered, and their attitudes toward features like progress visualization and spending alerts.

- **Q9. Are you currently trying to save for something specific?** (Single choice)
Yes, I have a precise goal / Yes, I have more than one goal / I save in general, without a specific goal / No, I'm not saving at the moment
- **Q9b. (Conditional) What are you saving for?** (Multiple choice)
Shown only if Q9 indicates a specific goal.
An important purchase / A trip or vacation / An emergency fund / Rent or housing expenses / Education or courses / A personal project / Other
- **Q10. Has it ever happened that you gave up on a savings goal or failed to reach it?** (Single choice)
Yes, more than once / Yes, once / No, I have always reached my goals / I have never tried to save for a specific goal
- **Q10b. (Conditional) What do you think influenced your giving up the most?** (Multiple choice, max 2)
Shown only if Q10 indicates a past failure.
Unexpected expenses absorbed the budget / I lost motivation over time / The goal was too distant or unrealistic / I couldn't tell whether I was making progress / Daily spending eroded my savings without me noticing / Other
- **Q11. How much do you agree with the following statements?** (Likert grid 1–5)
 - “Small daily expenses often feel irrelevant to me at the moment I make them.”
 - “I would find it useful to clearly see how much is left to reach a savings goal.”
 - “If I could receive an alert when I am spending too much relative to my goals, I would find it useful.”

- “I prefer to have manual control over my finances rather than relying on automatic tools.”

Section 4 — Toward a new tool

willingness to use an app with manual expense entry, preferred features, and an attitude toward data privacy.

- **Q12. Imagine an app where you can record expenses manually in less than 30 seconds, every time you make a purchase. How willing would you be to use it consistently?** (Likert 1–5)
1 = I would not use it, 5 = I would use it every day
- **Q13. Which of these features would you find most useful in a personal savings app?** (Multiple choice, max 2)
Creating multiple digital piggy banks, one per goal / Seeing charts and progress bars / Receiving alerts when my spending puts a goal at risk / Categorizing expenses to see where my money goes / Having an automatic monthly summary of my habits
- **Q14. How important is it to you that this kind of app keeps your data only on your phone, without sending it to external servers?** (Likert 1–5)
1 = I don't care, 5 = It is essential, I would not use an app that sends my data outside
- **Q15. Do you handle money in more than one currency?** (Single choice)
Yes, regularly / Occasionally (e.g. when traveling) / No, I use only Euros
- **Q16. Is there anything you would want from a savings app that current apps do not offer? Or any aspect of your financial habits that we have not touched on?** (Open, optional)

The questionnaire contained closed-ended questions (multiple choice, Likert scales, ranked options) and open-ended questions, following a conditional logic to skip irrelevant questions: for example, respondents who said they were not saving for a specific goal were not asked what they were saving for.

4.2.2 Sample Description

The sample turned out to be mainly of young adults (our target!): 63% of the interviewees fell into the 18–24 age group, and 29% were between 25 and 34 years old. No respondents fell in the 35–44 range, and only three were over 45 years old. In terms of employment, students and full-time employees were equally represented (13 each), with fewer freelancers (4) and part-time workers (3). The presence of full-time employees, while not the original primary target, helps us understand how the application could appeal to a slightly broader audience and confirms the relevance of our “young workers” angle.

Regarding income, 46% reported fixed and regular earnings, 17% described their income as fairly regular but with some variation, 26% reported irregular or variable income, and four respondents (about 11%) had no personal income at the time of the survey. Together, these four categories cover the entire sample.

Overall, we were fortunate; the target we were aiming for was the one with the highest response turnout, so we can move on to a coherent analysis of our results.

4.2.3 Results

Current Expense Tracking Behavior

One of the first things we wanted to understand was whether people actually keep track of their spending in any structured way. The results were quite telling: over half of the respondents (51%) reported that their only form of tracking is occasionally checking their bank account balance. Only one person out of 35 uses a dedicated budgeting app, and five respondents do not track their expenses at all.

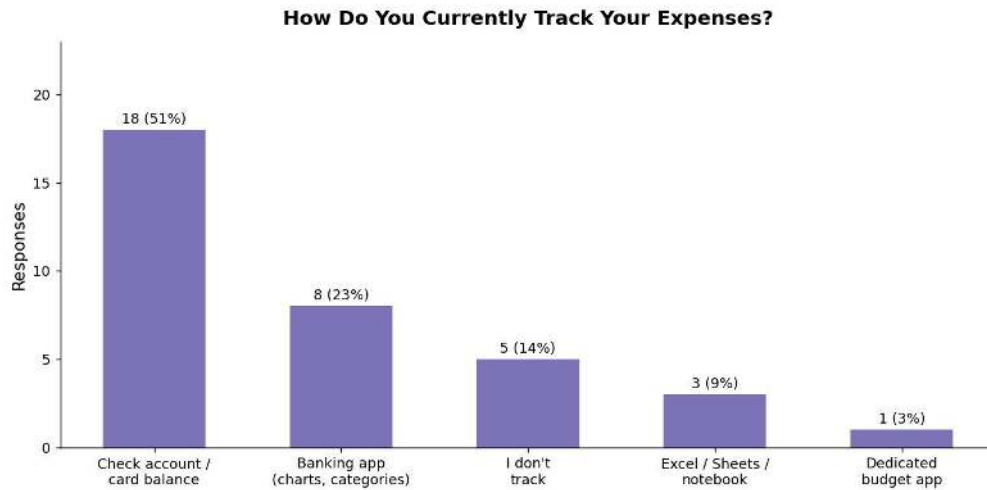


Figure 4.1: Methods currently used by respondents to track their expenses.

The challenge for our application therefore seems to be to offer users something new but simple, which can make the transition to active payment tracking less burdensome.

The Overspending Problem

The survey result that shocked us most concerned uncontrolled excessive spending at the end of the month: a full 77% of those interviewed said they had spent more than planned at least once.

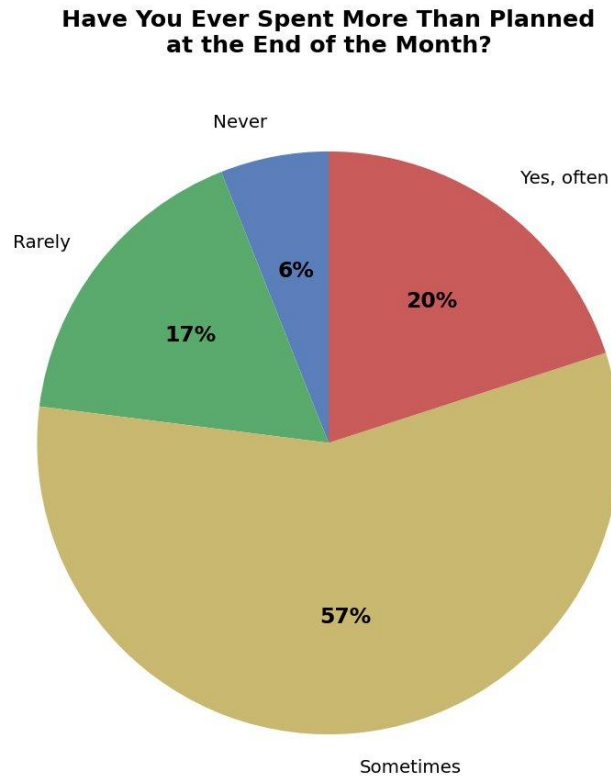


Figure 4.2: Respondents who have spent more than expected by the end of the month.

A very important discovery in our opinion, confirmed by the fact that small daily expenses, as shown in the graph, particularly dinners and aperitifs, are the least monitored and left forgotten.

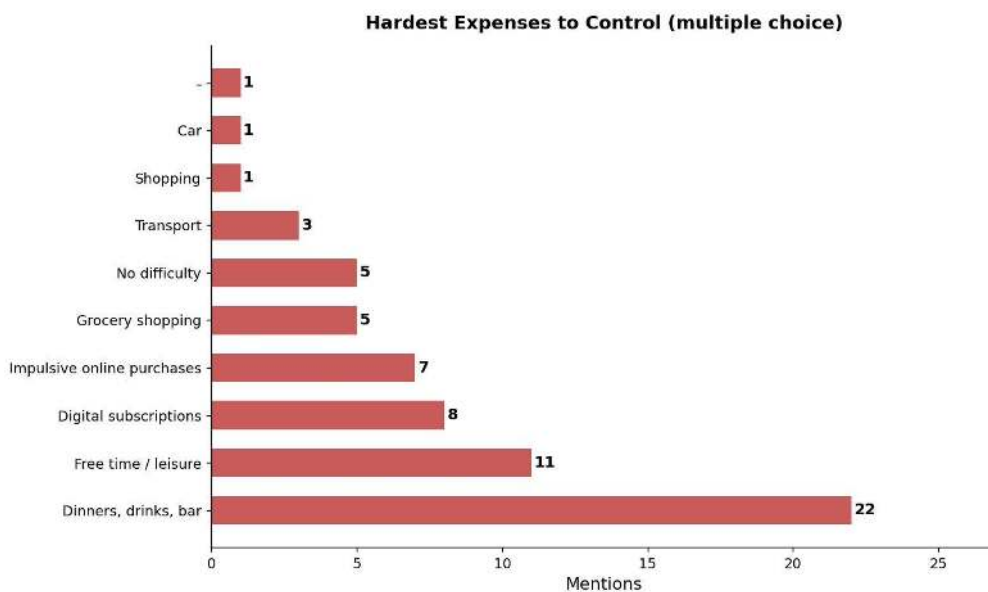


Figure 4.3: Expense categories that respondents find hardest to control (multiple answers allowed).

This pattern, in fact, aligns with the scenario of our persona 1 Chiara, who often spends on dinners and social activities without realizing the cumulative effect on her savings.

Savings Behavior and Obstacles

Regarding the survey on obstacles to saving, we found that approximately half of respondents (49%) have a specific savings goal in mind (such as a trip, etc.). On the other hand, a further 40% don't have a specific goal, saving just for the sake of it, to keep some money aside in case they need it in the future. This data gave us pause and helps us reiterate that the app must be designed with flexibility in mind, so that it can be useful even for those without a specific goal.

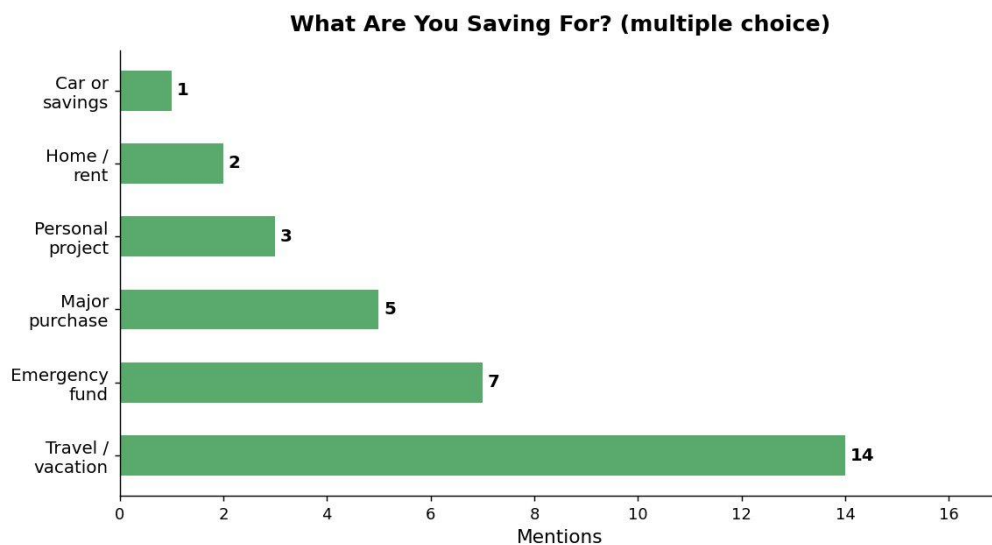


Figure 4.4: Savings objectives reported by respondents (multiple answers allowed).

Continuing our analysis of the results, we find another piece of data in our favor: among all those who gave up on a savings goal in their life, none did so due to a lack of motivation. This allows us to believe that the problem isn't so much a lack of willpower on the part of the user, but rather a lack of visibility into how daily habits influence medium- to long-term goals.

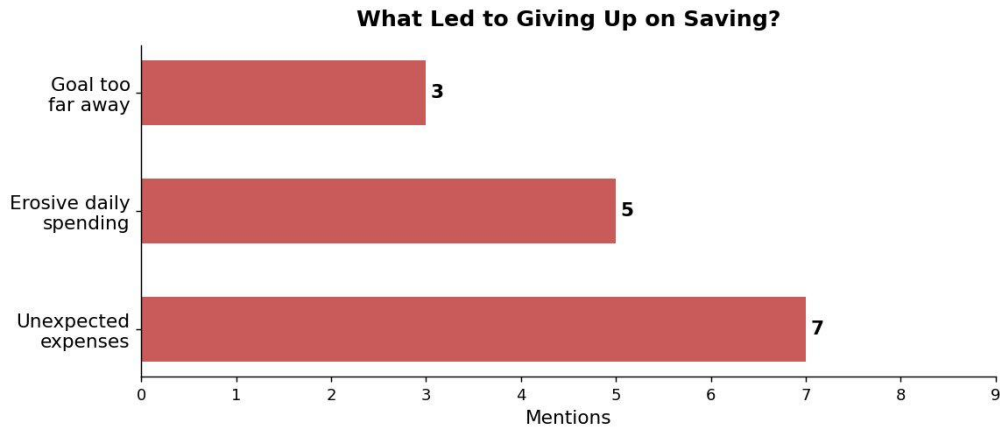


Figure 4.5: Reasons cited for giving up on a savings goal.

This result has a direct design implication: MyPiggyBank should prioritize features that make spending patterns visible and connect them to goal progress, rather than relying purely on motivational elements.

Attitudes Toward Key Features

Then we asked respondents to rate their agreement with four statements related to the core ideas behind MyPiggyBank, on a 1-to-5 Likert scale.

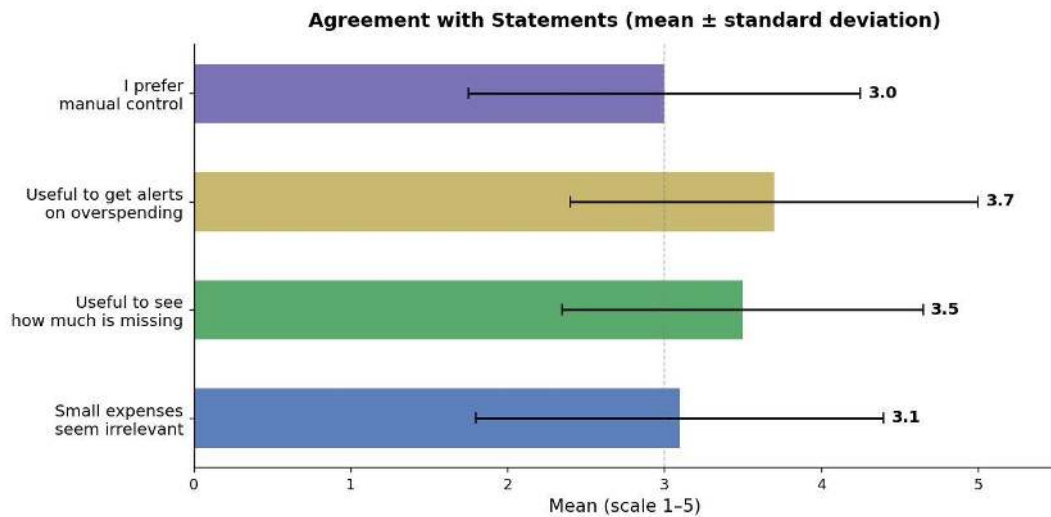


Figure 4.6: Average agreement with statements about spending awareness and tool preferences (1 = strongly disagree, 5 = strongly agree). Error bars represent one standard deviation.

The statement that manual control is preferred over automated tools scored exactly in the middle (average: 3.0/5), indicating that our user base is fairly divided on this issue. It's worth keeping in mind: although MyPiggyBank relies on manual input, the system shouldn't be too burdensome for users accustomed to automation.

Willingness to Use a Manual-Entry App

When presented with the idea of an app where expenses are recorded manually in under 30 seconds, the average willingness to use it consistently was **3.1 out of 5**. The distribution was fairly spread out, with 14 respondents scoring 4 or 5, but 10 scoring 1 or 2.

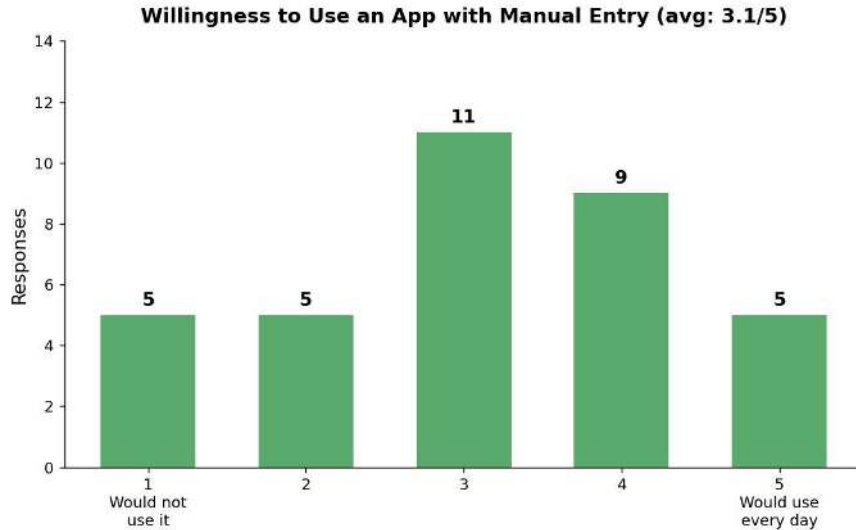


Figure 4.7: Willingness to use a manual-entry savings app on a daily basis.

We chose **not to hide** this honest result: the data tells us that manual entry is not a solution appreciated by everyone: it works for some users, but others are very skeptical. This reinforces a constraint we had already identified: entering expenses must be as quick and simple as possible, possibly achievable in two or three taps.

Feature Priorities

Respondents were also asked to select the two features they considered most useful in a personal savings app, so as to have a basis for understanding actual preferences within an ideal application. The monthly summary was the most requested feature, followed closely by the ability to create multiple digital piggy banks. Spending alerts and graphical progress bars, however, features we had initially considered central, received fewer requests.

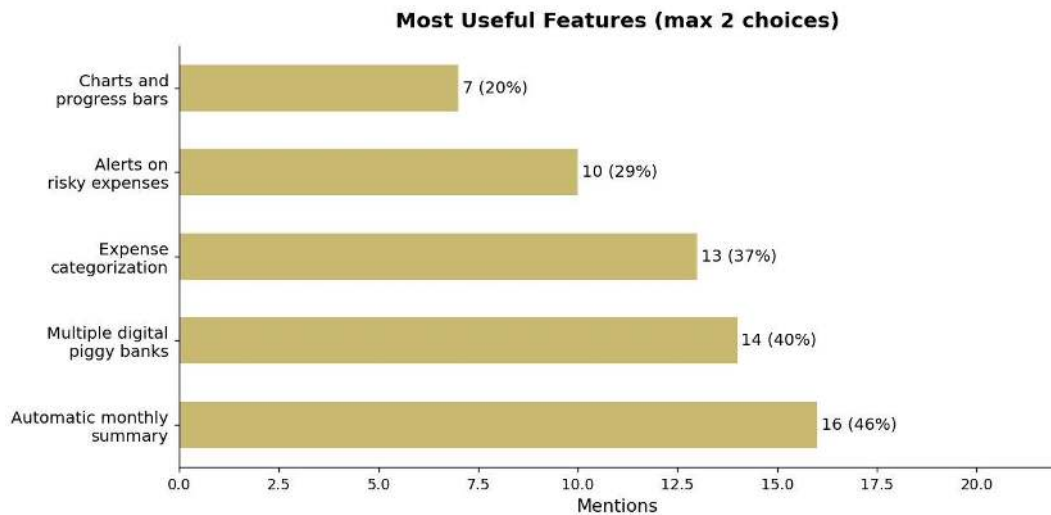


Figure 4.8: Features considered most useful by respondents (maximum 2 selections).

This was one of the most surprising findings. In the initial design, we had placed a lot of emphasis on visual progress indicators, but the survey suggests that users place greater importance on having a clear overview of where their money is going (categorization, monthly summaries) than on graphical elements like progress bars. This doesn't mean progress visualization should be removed, but it probably shouldn't be the dominant element of the interface.

Privacy

The issue of data privacy produced another equally important result. We see in the graph that the average score for the importance of keeping financial data exclusively on the user's device is 4.1 out of 5, with 21 out of 35 respondents selecting the highest possible value (5/5).



Figure 4.9: Importance of local-only data storage for financial information.

This strongly validates what we described in the application scenario: users are clearly uncomfortable with the idea of their financial data being sent to external servers.

Open-Ended Responses

Only four respondents provided additional comments in the optional final open-ended question. Although the sample is small, the suggestions are concrete, and we decided to highlight them:

- One respondent asked about the *ability to set spending limits by category*, with alerts when the threshold is approached.
- Another expressed interest in weekly reports with monthly forecasts, suggesting a need for *predictive functionality beyond simple expense tracking*.
- A third respondent proposed the option of *pre-loading fixed monthly expenses (rent, subscriptions, utilities) at the beginning of the month*, so that the remaining available budget is the actual amount that can actually be spent to avoid going into the red. This is a practical suggestion that could significantly improve the accuracy of savings tracking.
- The fourth comment noted that the topic of *personal investment* was not covered: this could be considered in future developments.

4.2.4 Key Takeaways

The results of the survey allow us to draw three conclusions that directly prepare us for the next phases of the project:

1. The idea works, but not for everyone. The 3.1/5 willingness to manually enter data tells us that half of our target audience will never use the app as we envisioned it. We can accept this or try to reduce the number of entries required (for example, by pre-loading fixed costs, as one respondent suggested).
2. The progress graphs, on which we had mentally built the app, are the least requested feature (7 votes versus 16 for the monthly summary). This result made us reconsider which feature should be most prominent on the home page, even though the actual redesign will be addressed in the design phase.
3. Regarding privacy, the data is clear and one-sided: 4.1/5 on average and 21 people out of 35 selecting the highest possible value. Local-first is no longer a design option; it's a real constraint.

In summary, the results of this questionnaire are reassuring as they confirm the validity of our underlying idea: daily expenses are a complex problem, which often erodes capital unnoticed. Therefore, taking into account the constraints listed above, we are ready to proceed with a new design phase.

4.3 Competitor Analysis

Before deciding exactly which features to develop, we examined those already available. Instead of just listing features, we tried to understand why these tools often fail to keep users engaged over time. We analyzed three very different approaches: a dedicated savings app, a large digital bank, and the classic ceramic piggy bank.

4.3.1 MoneyBox

MoneyBox is known for its “rounding up” mechanism: it automatically sets aside reserve change from every purchase. The user interface is clean, and the progress bars clearly show how close you are to a goal. However, during our group review, we noticed what felt like a major obstacle: the app forces you to connect a real bank account and constantly promotes paid investment products. If a user simply wants to track simple daily expenses to save for a new computer, this barrier to entry seems too burdensome and intrusive.

4.3.2 Revolut

A couple of us in the group have been using Revolut for years, so we’re very familiar with its “Vaults” and automatic rounding up. Revolut is incredibly fast: transactions are instant, and creating a vault takes literally seconds. The main issue, though? Automation actually undermines financial awareness. Because money moves automatically, you never really think about the act of saving. After a few months, you almost forget the vaults are there. Furthermore, Revolut only tracks what you spend with its specific card, completely neglecting cash or other accounts.

4.3.3 Physical Piggy Bank

We seriously considered the traditional piggy bank as a basic competitor. Physically dropping a coin into a slot creates a tangible connection with the act of saving that mobile apps struggle to replicate. Privacy is 100% guaranteed, and there are no technical barriers. The obvious downside is the complete lack of feedback: you don’t know how much you’ve saved until you open it, you don’t receive alerts if you’re overspending elsewhere, and it offers no data analytics.

4.3.4 Comparative Overview

From this comparison, two things become clear. First, no existing digital tool combines manual input (for awareness) with structured goal tracking and local-first privacy. Second, the old-school physical piggy bank actually scores very well on the things our users care about most: awareness and privacy. This might not apply to all users, but in our case, MyPiggyBank is basically our attempt to recreate that idea in a digital form, giving it the tracking and feedback capabilities it naturally lacks.

Feature	MoneyBox	Revolut	Piggy Bank	MyPiggyBank
Manual expense entry	×	×	×	✓
Multiple savings goals	✓	✓	×	✓
Progress visualization	✓	✓	×	✓
Spending alerts	×	Partial	×	✓
Expense categorization	×	✓	×	✓
Monthly summary	✓	✓	×	✓
Multi-currency support	×	✓	×	✓
Local-first / offline use	×	×	✓	✓
No bank account required	×	×	✓	✓
Spending awareness by design	×	×	✓	✓

Table 4.6: Feature comparison across competitors and MyPiggyBank.

4.4 Hierarchical Task Analysis (AS-IS)

The following Hierarchical Task Analysis describes how users **currently manage their personal savings without MyPiggyBank**, relying on the tools available to them today: physical containers (jars, envelopes), paper notebooks, spreadsheets, banking app dashboards, or mental tracking. The purpose of this AS-IS analysis is to identify the cognitive load, the steps most prone to error or omission, and the recurring pain points that the new system must address. The TO-BE re-decomposition, reflecting how the same tasks will be performed through MyPiggyBank, is implicitly addressed in Section 5 (State Transition Networks).

Following the methodology of Annett & Duncan, each top-level task is decomposed into sub-tasks expressed as actions, and each parent node is accompanied by an execution plan using the canonical plan types: *fixed sequence*, *optional*, *wait for event*, *cycle*, *time-sharing*, and *discretionary*.

Task 1: Set a savings goal

Goal. Define a financial target and plan how to accumulate the required funds without any dedicated software support.

Textual HTA.

- 1.1 Identify the good, service, or future need to save for.
- 1.2 Estimate the total amount required.
- 1.3 Choose a storage method (physical container, bank sub-account, paper notebook, spreadsheet).
- 1.4 Set a personal deadline for reaching the target.
- 1.5 Rank the new goal against existing financial commitments.

Plan 1. Do 1.1, then 1.2, then 1.3 in fixed sequence. If a deadline is relevant, do 1.4 (optional). If more than one goal is active, also do 1.5 (optional). *Type:* fixed sequence with optional steps.

Task 2: Record a daily expense

Goal. Track an outflow of money in order to preserve budget awareness during the month.

Textual HTA.

- 2.1 Complete the payment (cash or card).
- 2.2 Retain the information temporarily (keep the receipt or memorise the amount).
- 2.3 Retrieve the tracking medium (paper notebook, spreadsheet, mental tally).
- 2.4 Write down the amount spent.
- 2.5 Assign the expense to a mental or written category.
- 2.6 Recompute the remaining budget for the day or week.

Plan 2. Step 2.1 occurs whenever a transaction takes place (*wait for event*). Steps 2.2–2.5 are executed as a *cycle*, repeated for every individual expense. Step 2.6 is performed at the end of the recording session in *fixed sequence* after the cycle.

Task 3: Verify savings progress

Goal. Check how close the user is to the predefined target.

Textual HTA.

- 3.1 Inspect the storage medium (count cash, open the notebook, look up the bank balance).
- 3.2 Sum the amounts accumulated so far.
- 3.3 Compare the total against the target set in Task 1.
- 3.4 Compute the remaining amount needed.
- 3.5 Project a completion date based on the recent saving pace.

Plan 3. Do 3.1, then 3.2, then 3.3 in fixed sequence. Steps 3.4 and 3.5 are *discretionary*: the user performs them only when consciously planning, and frequently skips them because they require mental calculations.

Task 4: Adjust the savings strategy

Goal. React when current spending behaviour threatens one or more goals.

Textual HTA.

- 4.1 Recognise that something is off (usually by noticing a low balance or an unexpected expense).
- 4.2 Re-rank active goals by urgency.
- 4.3 Revise target amounts or deadlines.
- 4.4 Reduce planned spending in specific categories.

Plan 4. Step 4.1 is the trigger (*wait for event*: occurs only when the user notices a problem). Steps 4.2–4.4 are *discretionary* and combined freely depending on the situation.

4.5 Conclusions of the Analysis

The requirements analysis phase (combining user profiling, persona definition, user survey, competitor analysis, and hierarchical task analysis) has produced a set of findings that both confirm and refine the initial assumptions of the project.

Refined User Profile

The survey results largely validated the user profile defined at the start. The respondent base (63% aged 18–24, 29% aged 25–34) and the split between students and workers match the intended target. Two adjustments are worth noting. First, the original profile listed a motivation driven by “visual milestones”, but the survey showed that users actually prioritize monthly summaries and expense categorization over graphical progress indicators. The user profile (Table 4.1) has therefore been updated to describe a motivation driven by clarity and overview rather than by visual milestones alone. Second, the strong presence of full-time employees in the sample (13 out of 35) suggests that the target should explicitly include young salaried workers, not only freelancers and students; the User Profile has been adjusted accordingly.

Validated Design Assumptions

Three design assumptions received strong empirical support:

1. **The overspending problem is real and widespread.** 77% of respondents reported spending more than planned at least occasionally, confirming the core problem behind MyPiggyBank. The main culprits (dining out (22 mentions), leisure (11), and digital subscriptions (8)) are exactly the kind of small, recurring expenses that the application targets.
2. **Privacy is a hard requirement.** With an average importance score of 4.1/5 and 60% of respondents selecting the maximum value, local-first data storage is not a nice-to-have but a prerequisite. This validates the design constraint defined in the Application Scenario.
3. **The failure to save is a visibility problem.** Among those who gave up on a savings goal, the top reasons were unexpected expenses (7 mentions) and everyday spending eroding savings unnoticed (5 mentions). Nobody cited loss of motivation. This confirms that the system should focus on making financial behavior visible, not on gamification or motivational tricks.

Design Adjustments Based on Survey Findings

Two areas require adjustments compared to the initial design direction:

- **Feature prioritization.** The survey ranked monthly summary (16 mentions), multiple piggy banks (14), and expense categorization (13) as more important than visual progress bars (7). While progress visualization remains a useful element, the interface should give more prominence to summaries and categorization.
- **Manual entry must be extremely fast.** The average willingness to use a manual-entry app was 3.1/5. This means the expense entry flow is the single most critical

interaction to get right. If it takes more than a few taps, adoption will suffer. The open-ended suggestion to pre-load fixed monthly expenses could help reduce the number of entries required, and the HTA for Task 2 has already been updated to make currency selection optional with a sensible default.

Refined Functional Requirements

Based on the combined results, the core functionalities for MyPiggyBank are:

1. Creation of multiple saving goals with name, target amount, priority, and optional deadline.
2. Possibility to track each goal on a **weekly or monthly basis**, in line with the need expressed in the questionnaire for more frequent monitoring of progress.
3. Fast manual expense entry (under 30 seconds), with category selection and an optional currency override (default: user's preferred currency).
4. Monthly and weekly spending summaries with expense breakdown by category.
5. Pre-loading of fixed recurring expenses (rent, subscriptions, utilities) at the beginning of the month, so that the remaining budget reflects only discretionary spending.
6. Spending alerts triggered when behavior puts a saving goal at risk.
7. Visual progress indicators for each piggy bank, kept as a complementary element rather than the dominant one on the home screen.
8. Local-only data storage with no dependency on external servers.
9. Multi-currency support for users who occasionally spend in foreign currencies.

These requirements, together with the task decomposition from the HTA, will guide the design phase, starting with the definition of the State Transition Networks for the main interaction flows and the development of Prototype Zero.

Chapter 5

Design and Implementation

This chapter describes the dialogue design of MyPiggyBank using **State Transition Networks** (STNs), a diagrammatic notation that models the interaction between user and system as a set of states connected by arcs representing user actions and system responses.

Following the principle of **hierarchical STNs**, we first present a top-level network that shows the global navigation structure of the application. Each node at this level corresponds to a major functional area, and is then expanded into its own detailed sub-network. This approach keeps each individual diagram readable while still providing a complete specification of the interaction.

The nine refined functional requirements identified at the end of the analysis phase (Section 4.5) are mapped onto five main interaction areas: *Home Dashboard*, *Expense Entry*, *Piggy Bank Management*, *Reports and Summaries*, and *Settings*. All arc labels follow the convention: the top line states the **user action** (or system event), and the bottom line (in italics) states the **system response**.

5.1 Top-Level Navigation STN

The top-level STN shown in Figure 5.1 represents the global navigation structure of MyPiggyBank. After the app launch, the user lands on the *Home Dashboard*, from which every major functional area is reachable in a single tap, ensuring high **reachability**. Each sub-dialogue returns to the Home Dashboard upon completion or cancellation, guaranteeing full **reversibility** at the navigation level.

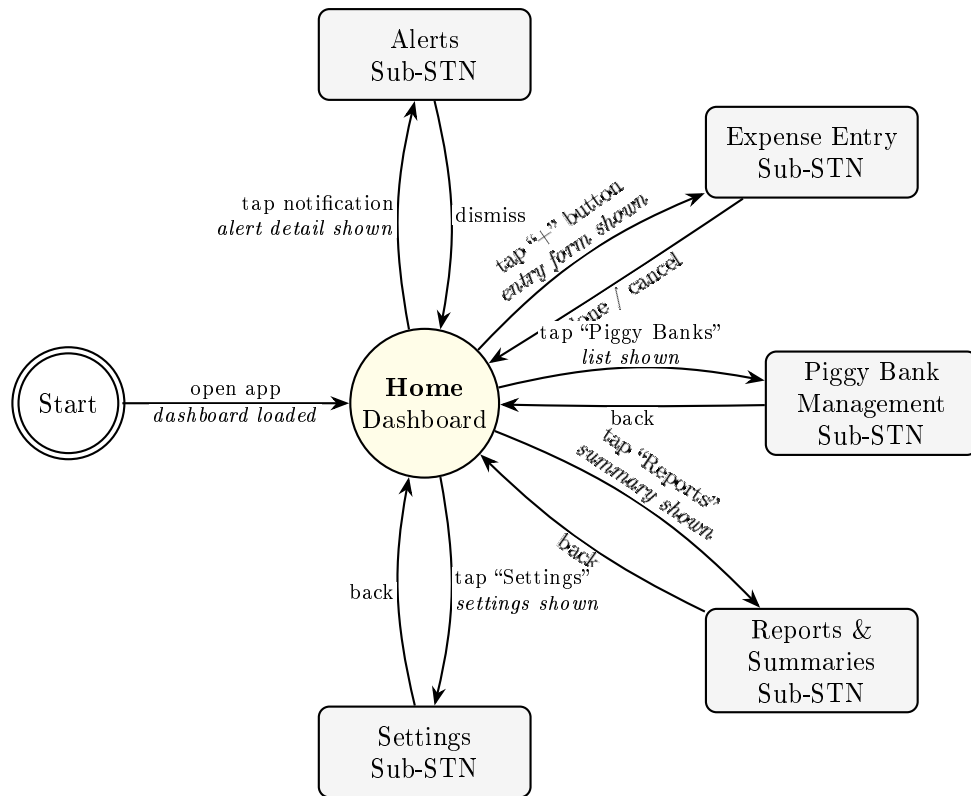


Figure 5.1: Top-level hierarchical STN of MyPiggyBank. Each rectangular node represents a sub-dialogue expanded in the following sections.

The bottom navigation bar of the app makes the Home Dashboard, Piggy Banks, Reports, and Settings tabs accessible from any state, which corresponds to a set of “escape” arcs active everywhere inside each sub-dialogue, in the same way described in the lecture material for handling global navigation shortcuts.

5.2 Sub-STN 1: Expense Entry

The Expense Entry flow is the most critical interaction in MyPiggyBank: as the survey revealed (Section 4.2.3), user willingness to adopt manual entry stands at 3.1/5, so the entire flow must be completable in under 30 seconds. The STN in Figure 5.2 was designed to require a minimum of two taps (amount + confirm) and a maximum of four (amount + category + currency + confirm), fulfilling the speed constraint from the Application Scenario.

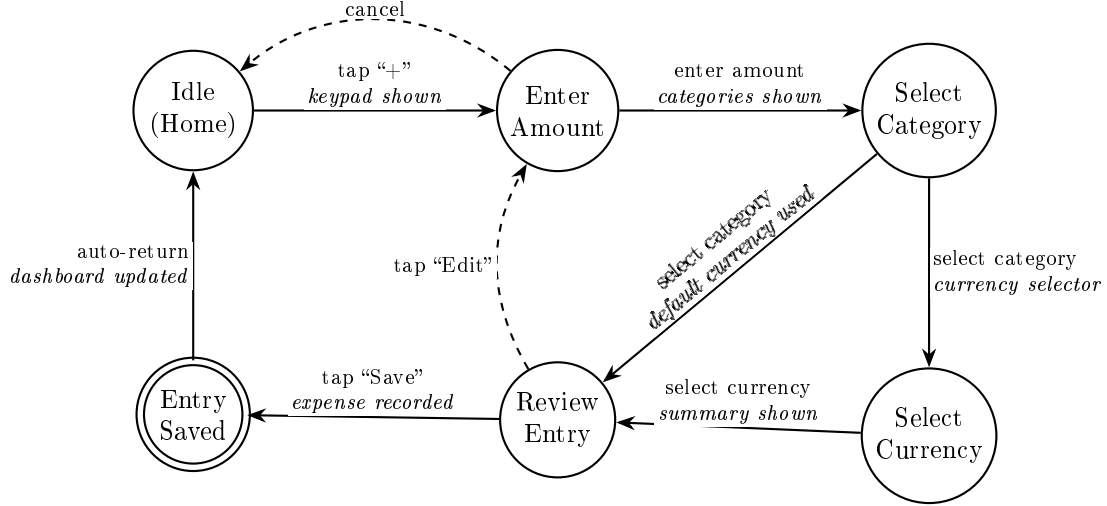


Figure 5.2: Sub-STN for Expense Entry. The dashed arcs represent escape/cancel paths. The shortcut arc from “Select Category” to “Review Entry” skips the currency selector when the user’s default currency is applied automatically.

Design decisions.

- The **default currency** shortcut avoids an unnecessary tap for 83% of users who reported using only euros (Q15 of the questionnaire). Multi-currency users can still override via the explicit currency selector.
- The “Edit” arc from *Review Entry* back to *Enter Amount* supports **reversibility**: the user can correct any field without losing the data already entered.
- After saving, the app **auto-returns** to the Home Dashboard in 1.5 seconds with an animated confirmation, so the user does not need to navigate back manually.
- **Error Handling and Edge Cases**: System-error states, such as invalid numeric inputs (e.g., entering alphabetic characters in the amount field) or database write failures, are intercepted at the “Enter Amount” and “Review Entry” nodes. The system displays a localized warning banner and prevents transition to the subsequent state until the input is successfully validated.
- **Empirical Speed Validation (Keystroke-Level Model)**: To rigorously validate the under-30-seconds constraint, we applied the Keystroke-Level Model (KLM) to the STN. For a default expense entry (amount + confirm), the sequence is: Mental preparation ($M \approx 1.2s$) + Pointing to amount field ($P \approx 1.1s$) + Tapping 4 digits ($4K \approx 0.8s$) + Pointing to Save ($P \approx 1.1s$) + Tapping Save ($K \approx 0.2s$) + System Response ($R \approx 1.5s$). The total execution time is $T_{execute} \approx 5.9$ seconds, which mathematically guarantees the system operates well within the 30-second threshold.

5.3 Sub-STN 2: Piggy Bank Management

The Piggy Bank Management dialogue covers the creation, editing, and deletion of savings goals, the second most requested feature in the survey (14 mentions, 40%). The STN in Figure 5.3 uses a hub-and-spoke structure centred on the *Piggy Bank List*, from which each operation is accessible.

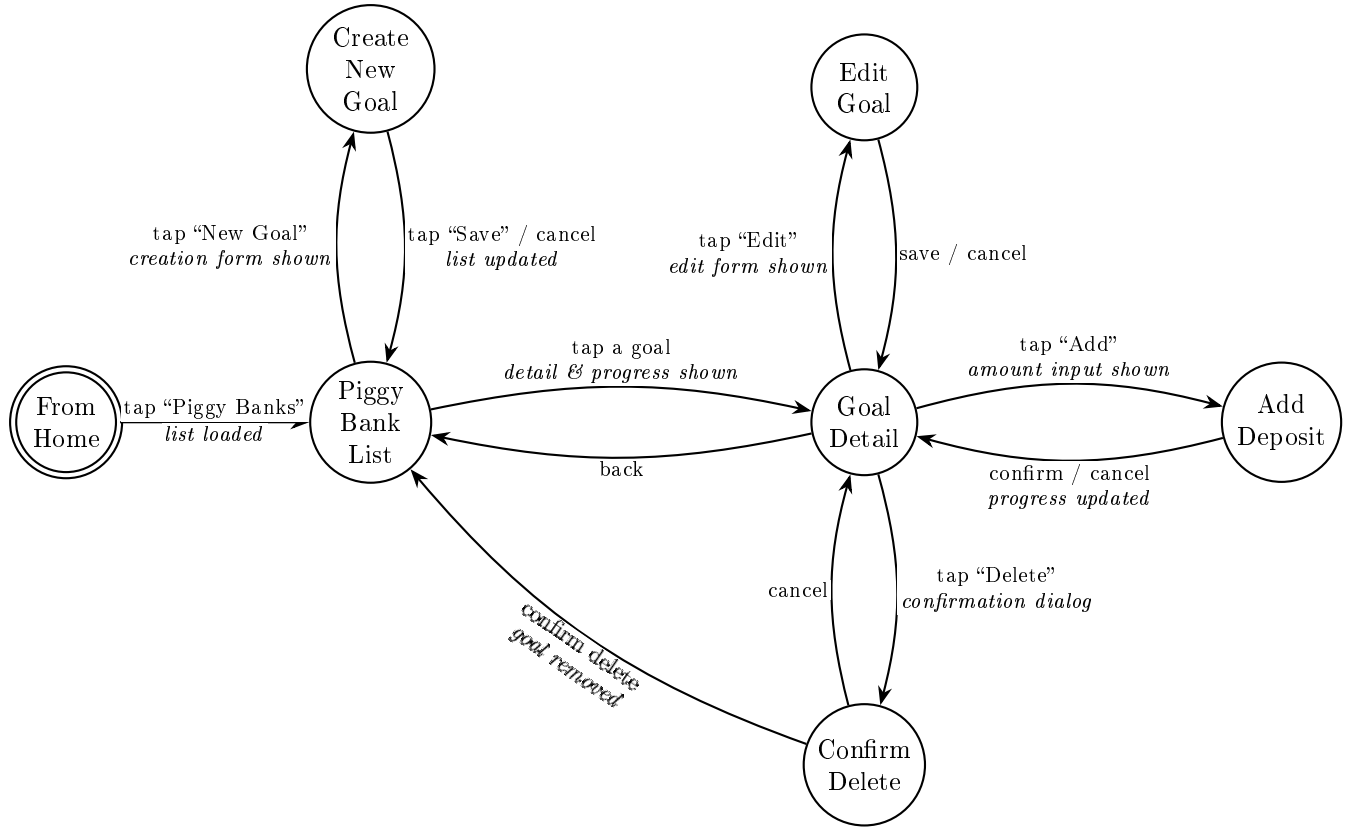


Figure 5.3: Sub-STN for Piggy Bank Management. The delete action passes through a confirmation state to prevent accidental data loss (protection from dangerous states).

Properties verified.

- **Reachability:** every state is reachable from the entry point in at most three transitions.
- **Reversibility:** every action is reversible within two steps (e.g., an accidental tap on "Delete" is reversed by tapping "Cancel" on the confirmation dialog).
- **Dangerous states:** the *Confirm Delete* state acts as a guard, requiring an explicit confirmation before the irreversible deletion of a savings goal.
- **Completeness:** every state handles the "back" event, so no user action leads to a dead end.

5.4 Sub-STN 3: Reports and Summaries

The Reports section responds to the top-ranked feature request: automatic monthly summaries (16 mentions, 46%). The STN in Figure 5.4 provides three views (weekly, monthly, and by-category) accessible from a single hub state.

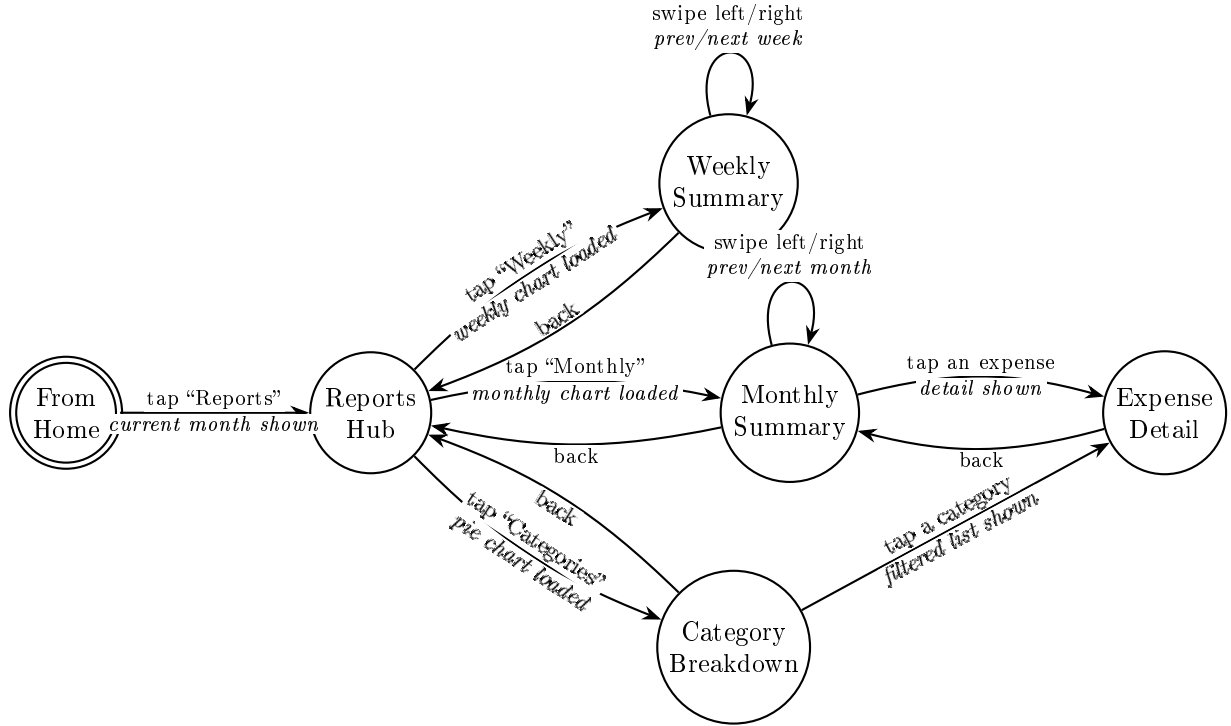


Figure 5.4: Sub-STN for Reports and Summaries. Swipe gestures on the weekly and monthly views allow period navigation without leaving the current state.

Design decisions.

- The **swipe gesture** for period navigation is modelled as a self-loop on the weekly and monthly states. This keeps the number of states manageable while allowing fluid temporal exploration.
- The **drill-down** from any summary view to an individual expense detail supports the survey finding that users want to understand where their money goes, not just see aggregate numbers.
- The Reports Hub defaults to the **current month** view, since the monthly summary was the most requested feature.

5.5 Sub-STN 4: Settings

The Settings dialogue covers user preferences, privacy controls, and multi-currency configuration. Because the survey showed that privacy is a hard constraint (average importance 4.1/5), the local data management options are given prominent placement.

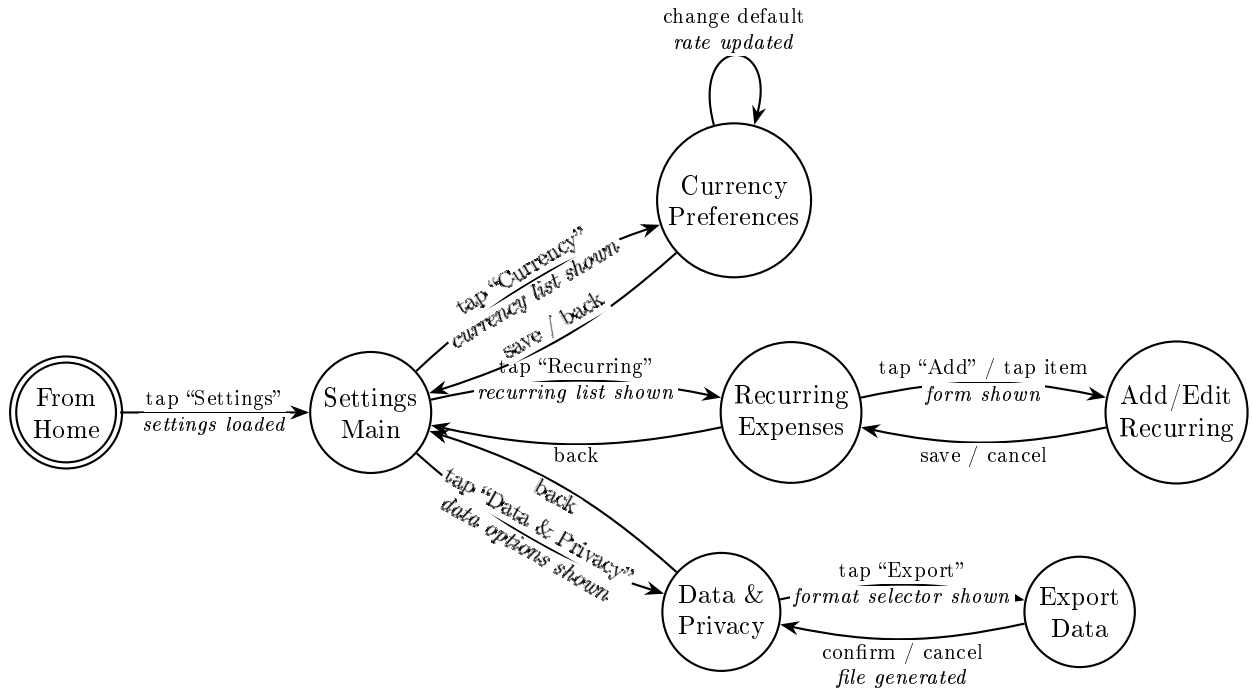


Figure 5.5: Sub-STN for Settings. The “Recurring Expenses” sub-area implements the open-ended survey suggestion to pre-load fixed monthly costs.

The **Recurring Expenses** feature directly implements a suggestion from the open-ended questionnaire responses: by pre-loading fixed costs (rent, subscriptions, utilities) at the beginning of each month, the remaining budget shown on the dashboard reflects only discretionary spending, significantly improving budget accuracy without requiring additional manual entries.

5.6 Sub-STN 5: Spending Alert Flow

Spending alerts are triggered by the system when the user’s behaviour puts a savings goal at risk. Unlike the other sub-dialogues, the alert flow begins with a **system-initiated event** (the alert notification) rather than a user action.

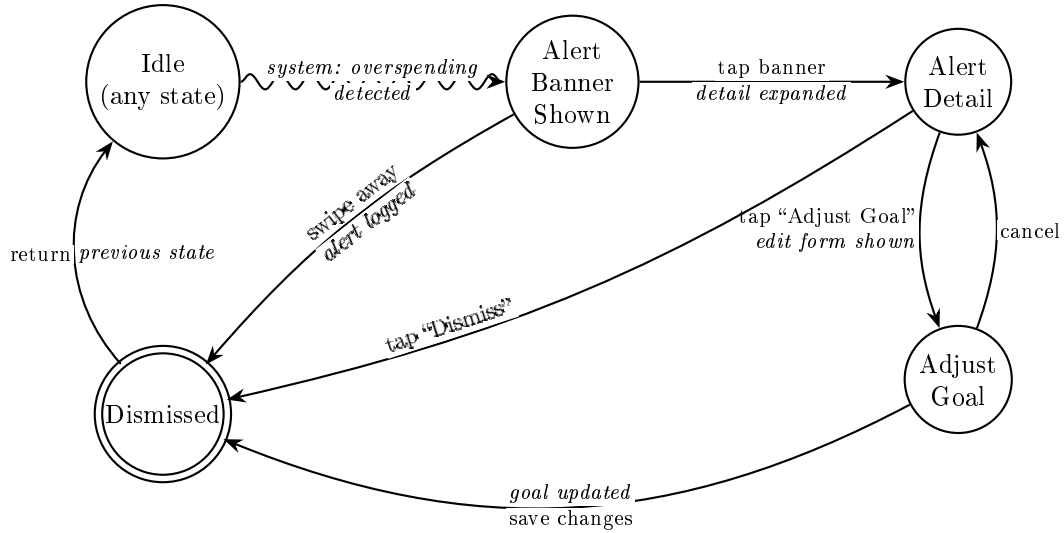


Figure 5.6: Sub-STN for the Spending Alert flow. The wavy arc indicates a system-initiated event. The alert is non-blocking: the user can dismiss it immediately or drill down to adjust the affected goal.

Design decisions.

- Alerts are **non-blocking**: a banner appears over the current screen, and the user can dismiss it with a swipe. This respects Chiara’s rejection of prescriptive notifications (Persona 1).
- The “Adjust Goal” shortcut lets the user immediately modify the target or deadline without navigating through the Piggy Bank Management flow, reducing friction.
- All dismissed alerts are logged in a notification history accessible from the Home Dashboard, so no information is lost.

5.7 Mapping to Functional Requirements

Table 5.1 traces each of the nine refined functional requirements (Section 4.5) to the specific STN sub-dialogue and states that implement it.

Functional Requirement	Sub-STN	Key States
FR1: Multiple saving goals	Piggy Bank Mgmt	Create New Goal, Goal Detail
FR2: Weekly/monthly tracking	Reports	Weekly Summary, Monthly Summary
FR3: Fast manual expense entry	Expense Entry	Enter Amount → Entry Saved
FR4: Spending summaries	Reports	Monthly Summary, Category Breakdown
FR5: Pre-loaded recurring expenses	Settings	Recurring Expenses, Add/Edit Recurring
FR6: Spending alerts	Alerts	Alert Banner, Adjust Goal
FR7: Visual progress indicators	Piggy Bank Mgmt	Goal Detail
FR8: Local-only data storage	Settings	Data & Privacy
FR9: Multi-currency support	Settings / Expense Entry	Currency Preferences, Select Currency

Table 5.1: Traceability matrix: functional requirements to STN sub-dialogues and states.

5.8 From HTA to STN: The TO-BE Perspective

The Hierarchical Task Analysis in Section 4.4 described how users currently perform savings-related tasks *without* MyPiggyBank (the AS-IS scenario). The STNs presented in this chapter represent the *TO-BE* re-decomposition: each HTA task is now mediated by the application, and the pain points identified in the AS-IS analysis are addressed as follows.

- **HTA Task 1 (Set a savings goal):** in the AS-IS, users must choose a storage method and mentally rank goals. In the TO-BE (Piggy Bank Management sub-STN), the app provides a structured creation form with name, target amount, priority, and optional deadline, and automatically ranks goals on the dashboard.
- **HTA Task 2 (Record a daily expense):** the AS-IS cycle of retain receipt → retrieve notebook → write amount → categorize → recompute budget is compressed into the Expense Entry sub-STN’s two-to-four-tap flow, eliminating the need for paper records.
- **HTA Task 3 (Verify savings progress):** the AS-IS manual sum-and-compare is replaced by the Reports sub-STN, which provides pre-computed weekly and monthly summaries with visual charts.
- **HTA Task 4 (Adjust strategy):** the AS-IS trigger (noticing a low balance) is now handled proactively by the Spending Alert sub-STN, which detects overspending patterns and notifies the user before the problem becomes critical.

This mapping confirms that the dialogue design covers all identified user tasks and systematically addresses the cognitive load, error-proneness, and omission risks documented in the AS-IS analysis.

Chapter 6

Prototypes

This chapter documents the prototyping phase of MyPiggyBank. Following the development methodology defined in the Work Plan (Chapter 3), we adopted an **evolutionary prototyping** approach: each prototype is a refinement of the previous one, and each iteration is paired with the evaluation technique most appropriate to its maturity.

Following the taxonomy of evaluation techniques presented in the course, we distinguish two families: *evaluation through expert analysis* (Heuristic Evaluation, Cognitive Walkthrough) and *evaluation through user participation* (Think Aloud, controlled experiment). The two families serve different moments of the design cycle. Expert analysis is cheap, fast, and ideally suited to early design, where it can surface a large share of usability problems before any user is involved; user-participation methods are more costly but provide evidence about how the system is actually used by real people.

Accordingly:

- **Prototype Zero** — a complete, navigable prototype covering all five sub-dialogues defined by the State Transition Networks (Chapter 5). Because it represents an early, full-coverage design, it was evaluated through *expert analysis*: a Heuristic Evaluation of the whole interface and a Cognitive Walkthrough of the single most critical task.
- **Prototype One** — the revised build that resolves the issues surfaced by the expert evaluation. Being a more mature artefact, it was submitted to *user-participation* evaluation: a qualitative Think-Aloud study and a controlled experiment, both reported in Section 6.2.

For each prototype we provide a brief description, the evaluation conducted (or designed), the results obtained, and the characteristics that distinguish each iteration from the next.

A short methodological note is in order. Prototype Zero was not produced as a set of hand-drawn wireframes; the five sub-STNs were translated directly into a working, clickable build, and the mockups reported below are screenshots of that build organised by sub-dialogue. This let the evaluators inspect realistic states — live progress bars, validation banners, populated lists — rather than static sketches, which makes the heuristic inspection and the cognitive walkthrough considerably more reliable.

6.1 Prototype Zero

Prototype Zero implements the complete navigation structure specified by the top-level STN and the five sub-STNs. Every state of every sub-dialogue corresponds to a concrete interface, and every arc to a navigation action. The prototype is organised below by sub-dialogue: each figure shows the mockups for one sub-STN, the navigation arrows connecting them (labelled with the triggering user action), and a short description of each interface’s purpose, as required for the heuristic evaluation.

6.1.1 Mockups and Interaction Flows

Top-Level Navigation — Home Dashboard

The Home Dashboard is the entry point of the application and the hub from which all sub-dialogues are reachable in a single tap. It surfaces the monthly budget consumption, the list of active piggy banks, and the two primary quick actions ($+ \textit{Expense}$ and $+ \textit{Goal}$). When the system detects that a savings goal has fallen behind its required pace, a non-blocking alert banner appears at the top of the screen, with a secondary entry point as a coloured card below the goal list.

Top-Level Navigation - Home Dashboard

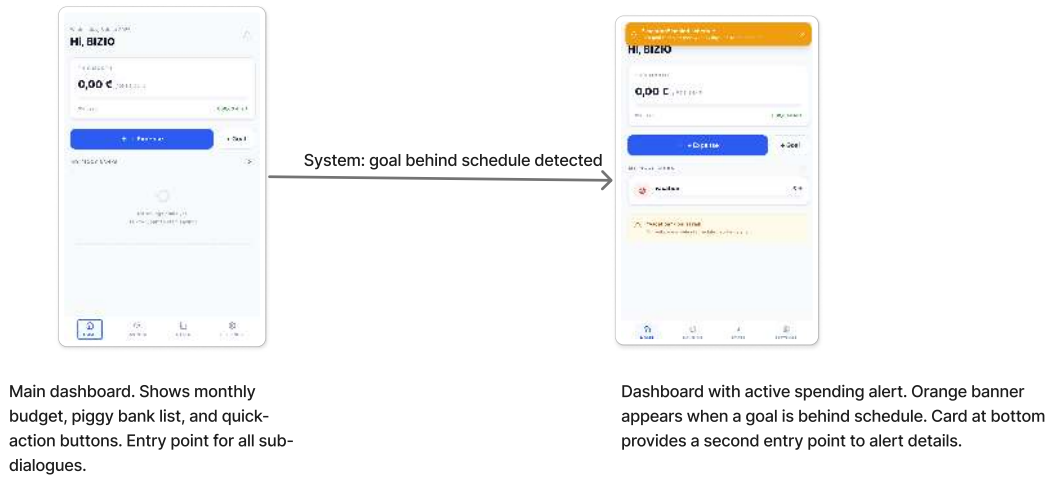


Figure 6.1: Top-Level Navigation. Left: the Home Dashboard in its normal state. Right: the same dashboard with an active spending alert (system-initiated). The arrow indicates the system transition *goal behind schedule detected*.

Sub-STN 1 — Expense Entry

The expense entry flow is the most critical interaction in the system: the user survey reported a willingness of only 3.1/5 to use a manual-entry app (Section 4.2.3), so the flow was designed to be completable in two to four taps. The sequence moves from amount

entry, through category selection (with a default-currency shortcut), to a review screen and a confirmation state that auto-returns to the dashboard. Invalid input is intercepted at the amount step with an inline error.

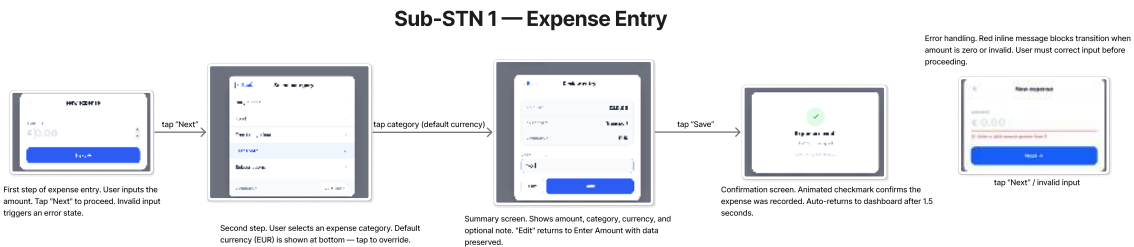


Figure 6.2: Sub-STN 1 — Expense Entry. Mockups and navigation flow from *Enter Amount* to *Entry Saved*, including the error-handling state (far right). Arrows are labelled with the user action that triggers each transition.

Sub-STN 2 — Piggy Bank Management

This sub-dialogue covers the creation, inspection, editing, and deletion of savings goals, the second most requested feature in the survey (14 mentions, 40%). It follows a hub-and-spoke structure centred on the Piggy Bank List, with a guarded confirmation dialog protecting the irreversible delete action.

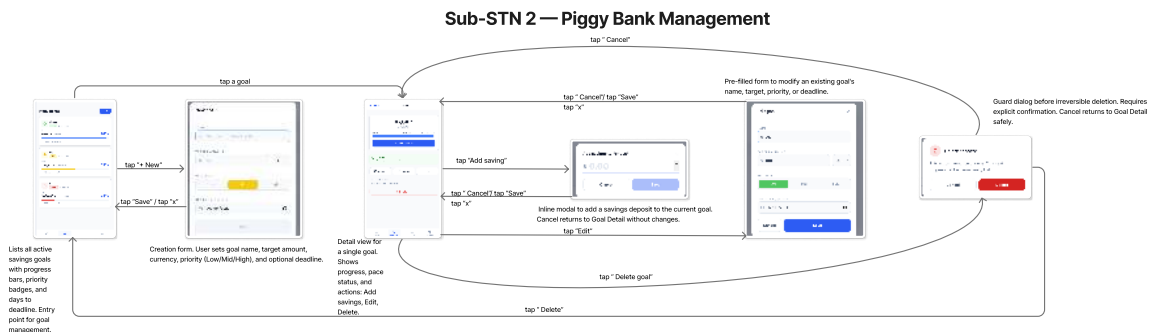


Figure 6.3: Sub-STN 2 — Piggy Bank Management. The flow covers the goal list, creation form, goal detail, add-deposit modal, edit form, and the delete confirmation guard. Cancel paths return the user safely to the previous state.

Sub-STN 3 — Reports and Summaries

The Reports section answers the top-ranked feature request (automatic monthly summaries, 16 mentions, 46%). It exposes three views — Monthly, Weekly, and Categories — from a single hub, with a drill-down to individual expense details.

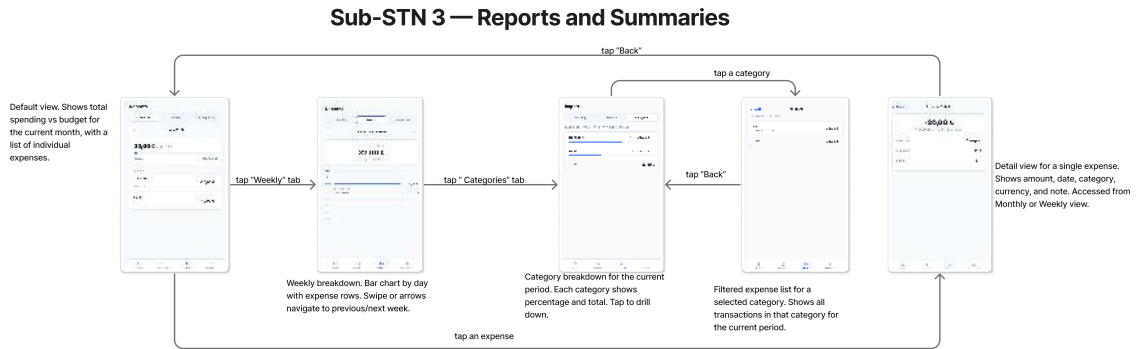


Figure 6.4: Sub-STN 3 — Reports and Summaries. The three tabbed views (Monthly, Weekly, Categories) and the drill-down to a single expense detail or a category-filtered list.

Sub-STN 4 — Settings

The Settings sub-dialogue groups profile, preferences, and privacy controls. Because privacy emerged as a hard constraint in the survey (average importance 4.1/5), the local-first storage notice and the data-export function are given prominent placement.

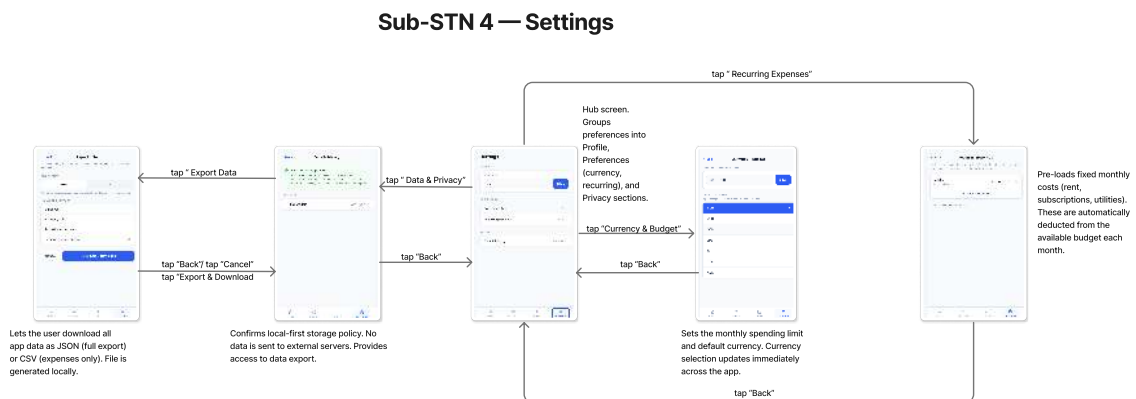


Figure 6.5: Sub-STN 4 — Settings. From the Settings hub to Currency & Budget, Recurring Expenses, Data & Privacy, and the Export Data screen.

Sub-STN 5 — Spending Alert Flow

The alert flow is the only sub-dialogue initiated by the system rather than the user. A non-blocking banner appears when the daily burn rate threatens a goal; the user can dismiss it (logging it to the notification history) or open it for detail.

Sub-STN 5 — Spending Alert Flow

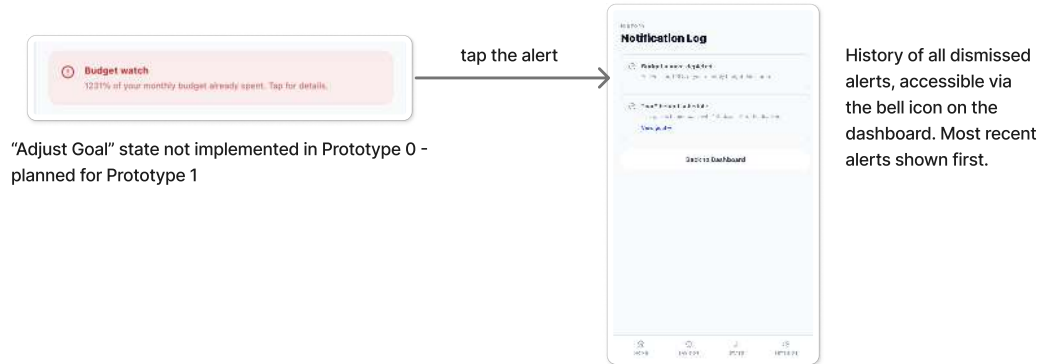


Figure 6.6: Sub-STN 5 — Spending Alert Flow. Tapping an alert card opens its detail and, on dismissal, records it in the Notification Log. The *Adjust Goal* state is discussed as a known gap below.

Known implementation gap. During the construction of Prototype Zero we identified one discrepancy between the Sub-STN 5 specification (Section 5.6) and the implemented build. The *Adjust Goal* state — the shortcut that should open the goal edit form directly from an alert — is not yet wired: tapping an alert card currently navigates to the Notification Log instead of opening the `GoalEditForm`. We documented this gap rather than hiding it, since surfacing exactly this kind of mismatch is one of the purposes of an early prototype. It was independently confirmed by the heuristic evaluation below (issue H3) and fixed in Prototype One.

6.1.2 Expert Evaluation

Once Prototype Zero was complete, we subjected it to an **evaluation through expert analysis**. As permitted by the course guidelines, we conducted *both* a Heuristic Evaluation (covering the whole interface) and a Cognitive Walkthrough (scoped to a single key task). The two methods are complementary: the heuristic evaluation gives broad coverage of interface-level usability problems, while the cognitive walkthrough deeply probes the learnability of the single most important task in the system.

The evaluation was carried out by the five members of the team acting as expert evaluators — Jacopo Rossi, Fabiano Cacioli, Emanuele Smisi, Fabrizio Pietrobono, and Luca Buonomini — on 25 May 2026. Each evaluator first inspected the prototype independently, after which the findings were consolidated. This independent-then-consolidate procedure follows the recommended practice, under which a small number of evaluators is expected to uncover the large majority of usability problems.

Heuristic Evaluation

The heuristic evaluation was structured around Jakob Nielsen’s ten usability heuristics. Each evaluator inspected the full set of mockups and interaction flows (Figures 6.1–6.6) and recorded each violation on a standard evaluation form together with the heuristic violated, the frame/page on which it appeared, and a severity rating on the 0–4 scale (0 = not a usability problem, 1 = cosmetic, 2 = minor, 3 = major, 4 = catastrophe). The complete form is reported in Appendix 10.1; the consolidated findings are summarised in Table 6.1.

#	Frame/Page	Heuristic violated	Description	Sev.
H1	Home Dashboard	H1 Visibility of system status H5 Error prevention	The budget-consumption figure could grow without bound (e.g. “1125005% used”, “9.000.040,00 €”) when an amount far above the budget was entered, with no sanity cap on the value or its display.	3
H2	Expense Entry; Goal detail	H3 User control & freedom H9 Recover from errors	A recorded expense cannot be edited or deleted afterwards. If the wrong amount is entered (see H1), the user has no recovery path.	3
H3	Spending Alert flow	H3 User control & freedom	Tapping an alert card does not lead to the expected <i>Adjust Goal</i> action; the user is taken to the Notification Log instead, with no direct way to act on the alert (confirms the known gap).	3
H4	New goal / Edit goal	H2 Match with real world	The deadline field displays the raw locale placeholder gg/mm/aaaa instead of a localized, human-readable date hint.	2
H5	Home Dashboard	H8 Aesthetic & minimalist design	The top alert banner and the “goal at risk” card below the list convey the same information twice, adding visual redundancy.	1
H6	Home Dashboard	H6 Recognition rather than recall	A newly recorded expense is not shown on the dashboard; the user must open Reports to see it. (<i>Not a requirement violation — recent expenses were never specified for the home screen — recorded as a design suggestion.</i>)	1

Table 6.1: Consolidated findings of the Heuristic Evaluation of Prototype Zero (25 May 2026), with Nielsen severity ratings (0–4). The full per-evaluator form is in Appendix 10.1.

The three highest-severity findings (all rated 3) were the unbounded budget display (H1), the impossibility of correcting a recorded expense (H2), and the missing *Adjust Goal* action (H3). Notably, H1 and H2 reinforce each other: an implausibly large entry is both poorly displayed *and* impossible to undo. These three issues, together with the localization and consistency problems, became the primary input to Prototype One.

Cognitive Walkthrough

The cognitive walkthrough was scoped, as required, to a single key task. We selected:

“Record a new expense of 20 € in the Transport category and save it.”

This task was chosen because Expense Entry is the most frequently performed and most adoption-critical interaction in MyPiggyBank: the survey willingness score of 3.1/5 means that any friction here directly threatens uptake. We first wrote the complete list of user actions and system responses, then, for each action, answered the four cognitive-walkthrough questions in the form used in the course:

1. Is the effect of the action the same as the user’s goal at that point? (*goal match*)
2. Will users see that the action is available? (*visibility*)
3. Once users have found the correct action, will they know it is the one they need? (*identification*)
4. After the action is taken, will users understand the feedback they get? (*feedback*)

The action–response list is reported below; the consolidated per-action verdict is summarised in Table 6.2, and the full four-question form is in Appendix 10.2.

- **Act. 1:** tap “+ Expense” on the dashboard. *Resp. 1:* the New Expense modal opens at the amount step.
- **Act. 2:** type “20” and tap “Next”. *Resp. 2:* the modal advances to the category list.
- **Act. 3:** tap “Transport”. *Resp. 3:* the category is ticked and, because the default currency applies, the modal advances directly to the review screen.
- **Act. 4:** tap “Save” on the review screen. *Resp. 4:* a confirmation screen with an animated checkmark is shown.
- **Act. 5:** (no action) the app auto-returns to the dashboard after 1.5 s.

Action	Q1	Q2	Q3	Q4	Note
1. Tap “+ Expense”	Yes	Yes	Yes	Yes	Largest, primary-coloured action on screen.
2. Enter 20, tap Next	Yes	Yes	Yes	Yes	Field focused on open; “Next” signals progression.
3. Tap “Transport”	Yes	Yes	Yes	<i>Partial</i>	Selecting a category also auto-advances (default-currency shortcut); a first-time user may momentarily expect a separate confirmation. The “default” currency label mitigates this.
4. Tap “Save”	Yes	Yes	Yes	Yes	Review restates amount/category/currency; “Edit” offers recovery.
5. Auto-return	Yes	—	—	Yes	Checkmark + “Returning to dashboard...” set expectation.

Table 6.2: Cognitive Walkthrough of the expense-entry task on Prototype Zero (25 May 2026).

The walkthrough confirmed that the expense-entry task is highly learnable: no blocking issue was found. The only documented negative was a partial answer to Q4 at Act. 3 — because tapping a category silently skips the currency step, a first-time user might not immediately understand that the flow has advanced. The explicit “default” label on the currency row was judged sufficient to prevent real confusion, so this was recorded as a minor observation rather than a defect. This positive result validated the central design bet of the project — that a two-to-four-tap manual entry can be made frictionless — and meant that the redesign effort for Prototype One could concentrate on the interface-level issues raised by the heuristic evaluation rather than on the core task itself.

6.1.3 From Prototype Zero to Prototype One

The expert evaluation produced a prioritized list of changes. The characteristics distinguishing Prototype Zero from Prototype One are summarised in Table 6.3.

Issue (source)	Change introduced in Prototype One
Unbounded budget figure (H1, sev. 3)	Added input sanity bounds and a display cap; the budget bar clamps and the percentage is rounded and bounded to a realistic range.
Expense not editable (H2, sev. 3)	Added the ability to edit and delete a recorded expense from its detail view, giving the user a recovery path from input errors.
Missing <i>Adjust Goal</i> action (H3, sev. 3)	Wired the alert card to open the <code>GoalEditForm</code> pre-filled with the affected goal, implementing the Sub-STN 5 shortcut as originally specified.
Localized date placeholder (H4, sev. 2)	Replaced the raw <code>gg/mm/aaaa</code> placeholder with a localized, human-readable date hint.
Redundant alert information (H5, sev. 1)	Consolidated the duplicate alert presentation on the dashboard into a single, non-redundant element.

Table 6.3: Key differences between Prototype Zero and Prototype One, traced to expert-evaluation findings.

6.2 Prototype One

Prototype One incorporates the five corrections listed in Table 6.3. The three major (severity 3) findings are all resolved: the budget figure is now bounded, a recorded expense can be edited or deleted, and the *Adjust Goal* shortcut is wired so that an alert leads directly to the affected goal’s edit form. The remaining localization and redundancy issues are also addressed.

6.2.1 Corrections Implemented

This section documents the implemented corrections, pairing each fix with the corresponding Prototype Zero state. Each figure shows the result in Prototype One and references the heuristic-evaluation issue it resolves.

Bounded budget figure (issue H1). In Prototype Zero the budget consumption could be displayed as an absurd value when an out-of-range amount was entered (Figure 6.7). Prototype One adds an input sanity bound at the amount step — amounts above 1,000,000 are rejected with an inline error (Figure 6.8) — and clamps the dashboard percentage to a realistic, rounded value.

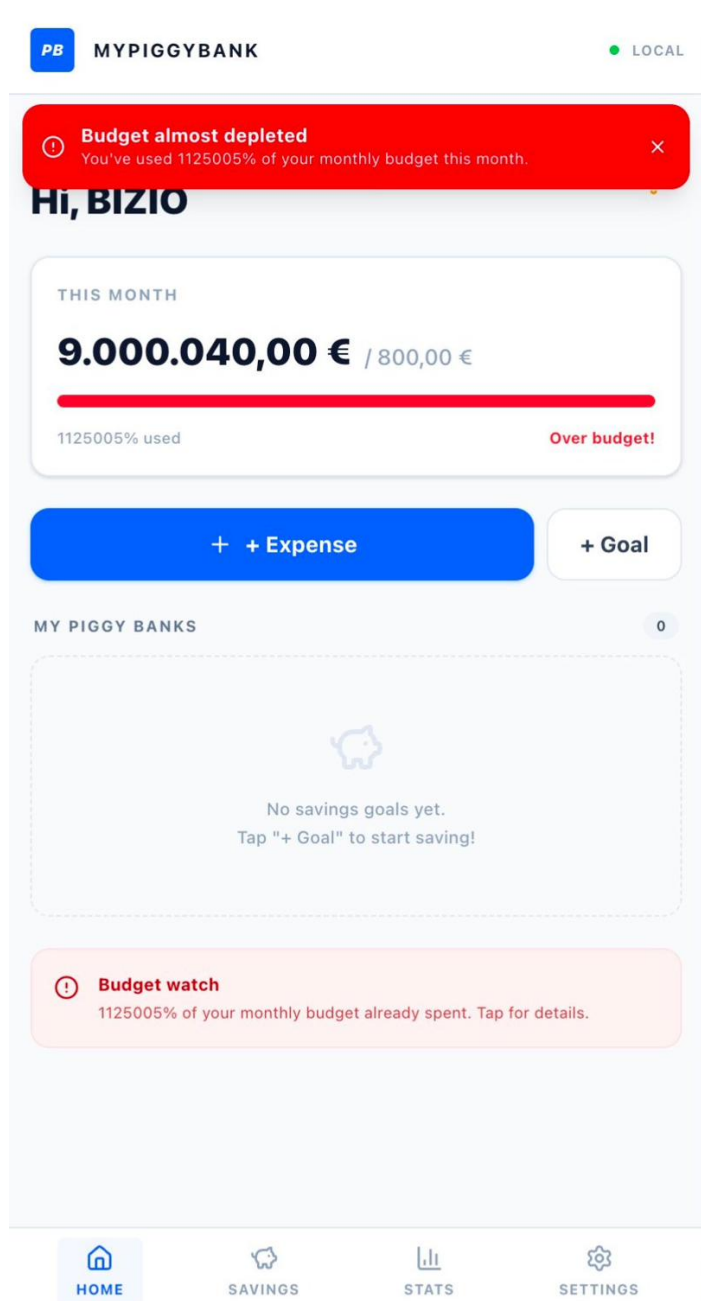


Figure 6.7: **Before** (Prototype 0): the budget shows “1125005% used” with no bound.

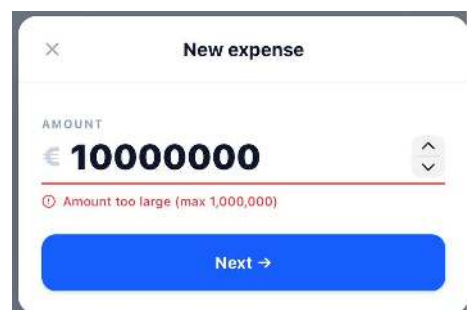


Figure 6.8: **After** (Prototype 1): out-of-range amounts are blocked with an inline error.

Editable and deletable expenses (issue H2). Prototype Zero offered no way to correct a recorded expense. Prototype One adds *Edit* and *Delete* actions to the expense detail (Figure 6.9), with deletion protected by a confirmation guard consistent with the goal-deletion dialog (Figure 6.10).

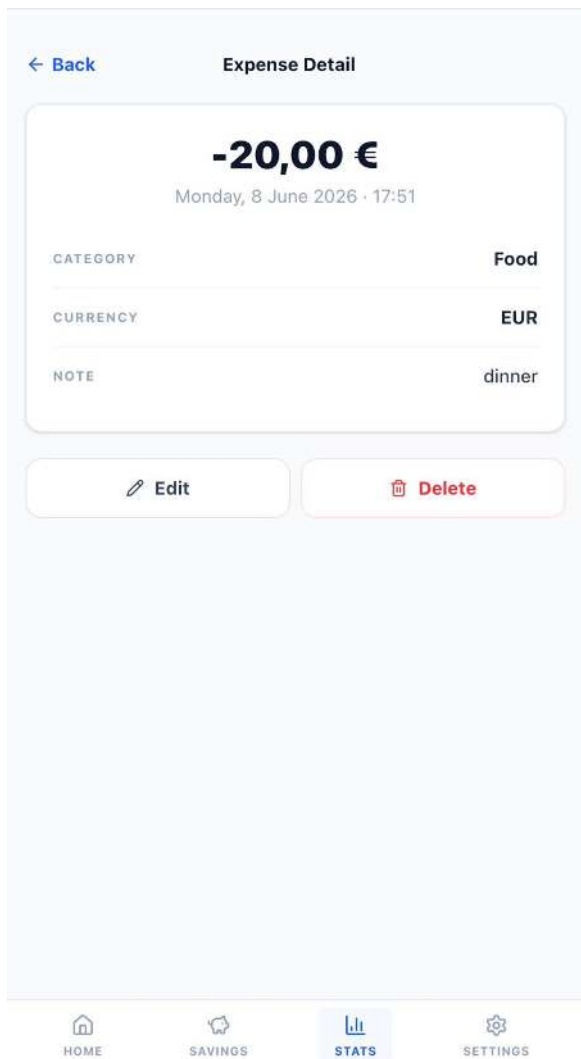


Figure 6.9: **After** (Prototype 1): the expense detail now exposes Edit and Delete.

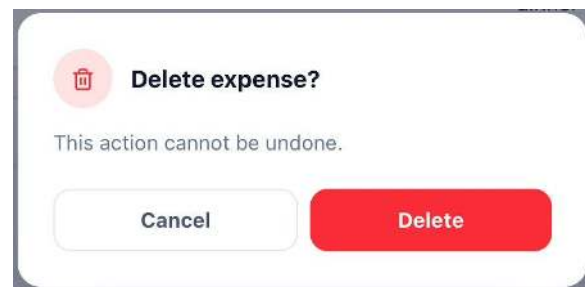


Figure 6.10: **After** (Prototype 1): deletion is protected by a confirmation guard.

Wired *Adjust Goal* shortcut (issue H3). The most important correction: in Prototype Zero, an alert card led to the Notification Log. In Prototype One, tapping a goal-at-risk alert opens its detail with an *Adjust Goal* action (Figure 6.11) that leads directly to the pre-filled goal edit form (Figure 6.12), implementing the Sub-STN 5 shortcut as originally specified.

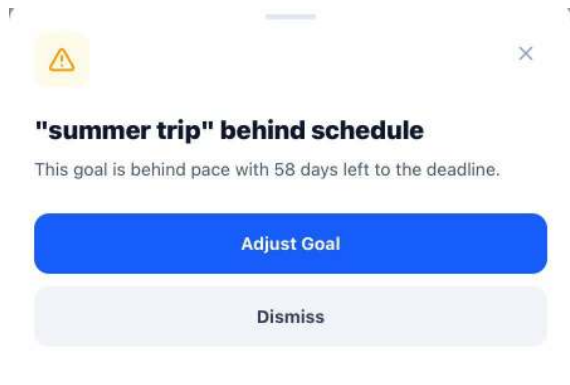


Figure 6.11: **After** (Prototype 1): the alert detail now offers *Adjust Goal*.

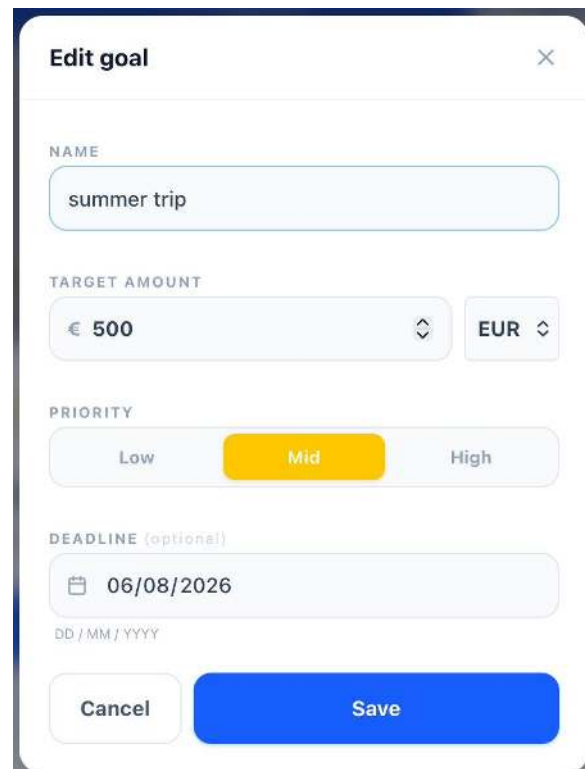


Figure 6.12: **After** (Prototype 1): *Adjust Goal* opens the goal edit form, pre-filled.

Localized date hint (issue H4) and consolidated alert (issue H5). The deadline field now shows a readable “DD / MM / YYYY” hint instead of the raw locale placeholder (Figure 6.13), and the dashboard presents the spending alert through a single element rather than a duplicated banner and card (Figure 6.14).

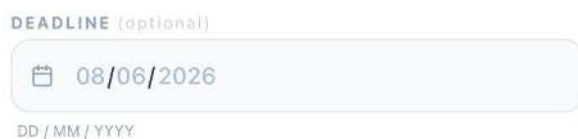


Figure 6.13: **After** (Prototype 1): the deadline field shows a readable date hint.

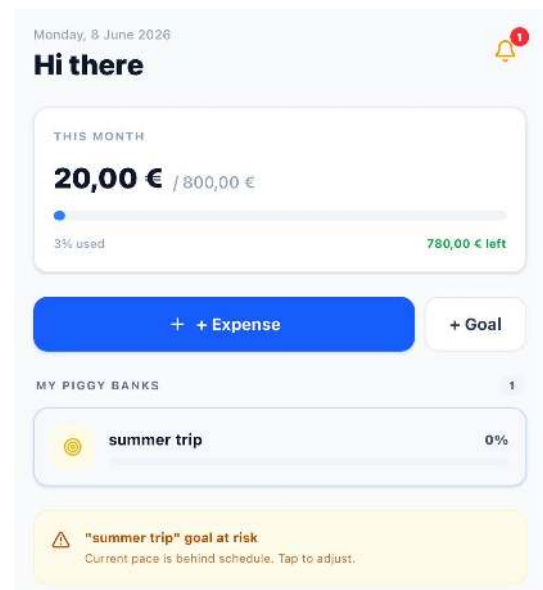


Figure 6.14: **After** (Prototype 1): a single, non-redundant alert presentation.

Having validated the core task through expert analysis, Prototype One was submitted to **evaluation through user participation**, comprising one qualitative assessment (a Think-Aloud study) and a controlled experiment. Both were carried out with real participants drawn from the target profile and from outside the development team.

6.2.2 Qualitative Evaluation: Think-Aloud Study

Method. The Think-Aloud method is simple, requires little equipment, and is well suited to revealing *how* the interface is actually understood. Each participant performed a task while continuously verbalising what they were doing, what they expected, and what they believed was happening; the moderator intervened minimally, only to remind the participant to keep talking. We recruited five participants matching the target profile (aged 18–34, mixed students and young workers, habitual smartphone users) — the established sweet spot for formative qualitative testing, beyond which the same problems tend to recur. Each was given three representative tasks mapped to the most important sub-dialogues: (1) record a new expense and assign it to a category; (2) create a new savings goal with a deadline and add a first deposit to it; (3) open the monthly report and find the single most expensive transaction. Following the course protocol, participants were told that the interface (not the person) was being tested, that they could stop at any time, and that the moderator could not help; hesitations, wrong turns, spontaneous comments, and completion were recorded.

Results. All five participants completed all three tasks. Task 1 (expense entry) was performed fluidly by everyone, with no hesitation, confirming on real users the learnability that the cognitive walkthrough had predicted. Task 3 (report navigation) was likewise completed by all, with individual times of 57, 47, 61, 55, and 49 seconds; the drill-down to the most expensive transaction was discovered without prompting. The two findings worth recording both concern Task 2 (goal creation and deposit):

- One participant mis-clicked while adding the first deposit, tapping “Add” on an existing goal instead of creating the deposit on the intended one. This pointed to a possible ambiguity in *where* the deposit action is initiated — the exact question the controlled experiment was designed to investigate.
- One participant hesitated on the goal *priority* selector and asked, in substance, how priority is actually handled: “if I set it to High, what really changes — just the colour?” This is a genuine usability observation: the priority control does not communicate its effect to the user. It is a finding the expert evaluation had not surfaced, which illustrates the complementary value of testing with real users.

One participant also volunteered that they liked the application. Overall, the study confirmed the strength of the core flows and surfaced one clear ambiguity (deposit initiation) that motivated the controlled experiment below, plus one smaller refinement target (communicating the meaning of goal priority).

6.2.3 Controlled Experiment

Question. The Think-Aloud mis-click raised a concrete design question that opinion alone could not settle: *where* should the “Add savings” deposit action live? **Design A** places a deposit button directly on each goal card in the list (one tap from the list),

whereas **design B** keeps the deposit button inside the goal detail screen only (requiring the user to open the goal first). We ran a controlled experiment, structured in the five steps presented in the course, to decide on measured behaviour rather than preference.

Step 1 — Hypothesis. H_1 : the placement of the deposit button changes the time needed to add a deposit. H_0 : there is no difference in mean task-completion time between design A and design B. The experiment aims to disprove H_0 .

Step 2 — Variables. The **independent variable** is the deposit-button placement (two levels: A, on the card; B, inside detail). The **dependent variables** are task-completion time (seconds) and number of mis-taps. The **controlled variables** are the task wording, the seed data, the device, and the starting screen, all held identical.

Step 3 — Method. A *between-groups* design: each participant used only one variant, eliminating any transfer of learning. Twelve participants matching the target profile were randomly split into two equal groups of six; the goal state was reset to the same starting point before each participant, and the same experimenter timed every run for consistency.

Step 4 — Run. Each participant performed the task “add a 50 € deposit to the *trip* goal” once under the assigned variant, with completion time and mis-taps recorded. The raw times are reported in Appendix 10.4.

Step 5 — Analysis. A one-way ANOVA at $\alpha = 0.05$ compared the two group means. The results are summarised in Table 6.4.

Source of variation	SS	df	MS	F
Between groups (A vs. B)	11.41	1	11.41	1.19
Within groups (error)	95.62	10	9.56	
Total	107.03	11		

Table 6.4: One-way ANOVA for the deposit-button placement experiment. Group A mean = 9.40 s; Group B mean = 11.35 s.

Result. The card-level variant A was faster on average (9.40 s vs. 11.35 s), but with $F = 1.19$ against $F_{\text{crit}}(1, 10) = 4.96$ and a p -value of $0.30 > 0.05$, the difference is **not statistically significant**. We therefore **fail to reject the null hypothesis**: the data do not support a claim that button placement affects completion time. The within-group variance (individual times ranged from about 6 to 16 seconds in both groups) is large relative to the roughly two-second gap between the means, so with six participants per group the observed difference is well within what chance could produce. On mis-taps, variant A recorded one (the same ambiguity seen in the Think-Aloud) against zero for variant B, but on such a small sample this difference carries no statistical weight.

Design decision. Because the experiment provides no significant evidence that the card-level button is faster, the rational choice is to **retain the simpler existing design (variant B)** rather than add a card-level control that is not demonstrably better. This

is the conclusion carried into the final realization (Chapter 7). The episode is a textbook illustration of the method working as intended: a qualitative observation generated a hypothesis, the hypothesis was tested rigorously, and the evidence told us *not* to make the change — a more informative and more honest outcome than a convenient confirmation would have been.

Chapter 7

Final Realization

The final realization of MyPiggyBank corresponds to Prototype One: the build that resolves all the issues raised by the expert evaluation and that is ready for the user-participation studies designed in Section 6.2. This chapter describes its features, its strengths and weaknesses, and the improvements made with respect to Prototype Zero.

7.1 Features

The product delivers the nine refined functional requirements defined in Section 4.5, organised across the five sub-dialogues:

- **Fast manual expense entry** (FR3) — a two-to-four-tap flow, validated at ≈ 6 seconds by the Keystroke-Level Model (Section 5.2) and found highly learnable by the cognitive walkthrough; now with editable/deletable entries for error recovery.
- **Multiple savings goals** (FR1) with name, target, priority, and optional deadline.
- **Weekly and monthly tracking** (FR2) and **spending summaries** (FR4) through the Reports hub, with drill-down to individual expenses.
- **Pre-loaded recurring expenses** (FR5), implementing the open-ended survey suggestion to deduct fixed monthly costs automatically.
- **Spending alerts** (FR6), now including the fully wired *Adjust Goal* shortcut from the alert detail.
- **Visual progress indicators** (FR7), kept as a complementary rather than dominant element, in line with the survey finding that summaries outrank progress bars.
- **Local-only data storage** (FR8) with JSON/CSV export, addressing the hard privacy constraint (4.1/5).
- **Multi-currency support** (FR9) with a local exchange-rate cache and an optional per-expense currency override.

7.2 Strengths

The product's clearest strength is a genuinely fast and learnable core task. Recording an expense — the most adoption-critical interaction in the whole system — passed every cognitive-walkthrough question, which directly addresses the central risk the survey had exposed, namely the lukewarm 3.1/5 willingness to adopt manual entry. Closely related

is the robustness of the error handling that emerged from the expert evaluation: the two reinforcing defects around erroneous entries, an unbounded display and the absence of an undo path, were both fixed, closing what had been a clear usability hole. Beyond the individual interactions, the architecture itself is a strength: the local-first approach is not a feature added at the end but a constraint honoured throughout, matching the strongest and least ambiguous signal in the survey. Finally, and perhaps most importantly for a course project, the work is honestly traceable end to end: every interface traces back to a functional requirement, every requirement to a survey finding, and every change in Prototype One to a specific, severity-rated expert-evaluation result.

7.3 Weaknesses

The most significant weakness concerns the scale of the user-participation evaluation. The Think-Aloud study (five participants) and the controlled experiment (twelve) were carried out, but these sample sizes, while appropriate for formative testing, are too small to support strong generalization; the controlled experiment in particular returned a non-significant result, which means no firm conclusion about deposit-button placement can be drawn from it. A second limitation is intrinsic to the chosen paradigm: manual entry is not universal. The survey’s willingness distribution was bimodal, with ten of thirty-five respondents scoring only 1 or 2, and for those users the manual approach itself — however fast it has been made — may remain a barrier that the pre-loading of fixed costs only partially mitigates. Accessibility is also still limited: as acknowledged in the Application Scenario, the interface is highly visual, and screen-reader support, dynamic type, and high-contrast modes have not yet been implemented. Lastly, multi-currency support depends on connectivity for rate freshness; conversions remain available offline through cached rates, but the accuracy of those rates degrades during long offline periods.

7.4 Improvements Over the Previous Version

Relative to Prototype Zero, the final build fixes all three major (severity 3) expert-evaluation findings — the unbounded budget figure, the inability to correct a recorded expense, and the missing *Adjust Goal* action — and resolves the localization and redundancy issues as well. The cumulative effect is a prototype whose every non-trivial change is backed by a documented, severity-rated usability finding rather than by intuition.

Chapter 8

Conclusions and Future Work

8.1 Key Points of the Realization

MyPiggyBank set out to recreate, in digital form, the mindful quality of the old ceramic piggy bank: deliberate manual entry, instant visual clarity, and absolute privacy. The user-centered process — from survey-driven requirements, through STN-based dialogue design, to an expert evaluation of the first full prototype — produced a build that delivers on that intention while remaining honest about what has and has not yet been tested.

The decisive moments of the realization were the moments where evidence overruled intuition. The survey told us that progress bars, which we had assumed central, mattered less than monthly summaries; we demoted them. The survey told us privacy was non-negotiable; we made local-first a constraint, not an option. The heuristic evaluation told us that fast entry without an undo path is a liability, not a feature; we added editing and deletion. In each case the HCI method protected the product from our own assumptions.

8.2 Critical Phases and How They Were Overcome

Three phases of the project were genuinely difficult and shaped the final result. The first was translating requirements into dialogue. Mapping nine functional requirements onto five coherent sub-dialogues was not mechanical, and it was the State Transition Network notation that made it tractable: by forcing every state and every transition to be made explicit, the STNs turned what could have been guesswork into a structured exercise, and as a useful side effect made the later enumeration of mockups follow directly from the diagrams. The second difficulty was an unresponsive expert channel. When external expert feedback did not arrive within the deadline, the team conducted the heuristic evaluation and the cognitive walkthrough internally, against Nielsen's heuristics and the four-question walkthrough protocol taught in the course; this kept the iteration on schedule and still surfaced three severity-3 issues that drove Prototype One. The third challenge was interpreting a non-significant experimental result. The controlled experiment on deposit-button placement did not reject the null hypothesis, and the temptation in that situation is to force a favourable reading of the data. We resisted it: rather than claim a speed advantage the statistics did not support, we accepted the null result and let it dictate the conservative design choice (keeping the simpler layout), which is both

the more rigorous and the more honest response.

8.3 Reflection on the HCI Techniques Used

Looking back, the techniques did not contribute equally. The State Transition Networks were by far the highest-leverage tool in the project: they structured the design, guaranteed properties such as reachability and reversibility, and made the prototype’s interface inventory derivable directly from the diagrams. The Keystroke-Level Model, though far smaller in scope, earned its place by giving a cheap, early, quantitative validation of the thirty-second speed constraint well before any user was involved. Among the evaluation methods, the heuristic evaluation proved the most efficient in cost/benefit terms — a single afternoon of independent inspection produced six documented issues, three of them major — while the cognitive walkthrough complemented it by scrutinising the single most critical task in depth. The user-participation methods earned their keep differently: the Think-Aloud surfaced a usability concern the expert analysis had missed (the unclear effect of goal priority), and the controlled experiment, while returning a non-significant result, proved the most instructive of all — it disciplined us into accepting that a plausible design change was not justified by the evidence, which is precisely the kind of result that distinguishes empirical evaluation from intuition. The competitor analysis, by contrast, contributed comparatively little new direction: it confirmed a market gap we had already inferred from the survey, but did not by itself change any design decision.

8.4 Future Work

The weaknesses identified in Chapter 7 map directly onto a future-work agenda, the first item of which is the most immediate:

- **Replicate the user-participation evaluation at larger scale.** The Think-Aloud (five participants) and the controlled experiment (twelve) were formative; a larger, more diverse sample would give the deposit-placement question the statistical power this iteration lacked, and would let further interface questions be settled empirically.
- **Accessibility.** Add screen-reader semantics, dynamic type, and high-contrast themes to move beyond the current visual-first interface.
- **Lowering the manual-entry barrier further.** For the skeptical 1–2 segment of the survey, explore optional semi-automatic capture (e.g. receipt scanning or quick-add widgets) that preserves the mindful character of the app without forcing every entry by hand.
- **Shared goals.** Persona Giulia’s need to coordinate couple goals (Section 4.1) remains unaddressed; a privacy-preserving shared-goal mechanism is a natural extension.
- **Predictive summaries.** One open-ended survey respondent asked for weekly reports with monthly forecasts; lightweight, on-device projection of the burn rate would answer this.

Chapter 9

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Chapter 10

Appendix

In keeping with the principle of a lean main document, this appendix collects the full evaluation forms and protocols. The main text (Chapter 6) reports the consolidated analysis; the detailed, repetitive records are gathered here for further study.

10.1 Heuristic Evaluation — Full Form

Evaluators: Jacopo Rossi, Fabiano Cacioli, Emanuele Smisi, Fabrizio Pietrobono, Luca Buonomini **Prototype:** Zero **Date:** 25 May 2026

Severity scale (Nielsen).

- **0** — I don't agree that this is a usability problem at all.
- **1** — Cosmetic problem only; fix if extra time is available.
- **2** — Minor usability problem; low priority.
- **3** — Major usability problem; high priority to fix.
- **4** — Usability catastrophe; imperative to fix.

The ten heuristics used. (1) Visibility of system status; (2) Match between the system and the real world; (3) User control and freedom; (4) Consistency and standards; (5) Error prevention; (6) Recognition rather than recall; (7) Flexibility and efficiency of use; (8) Aesthetic and minimalist design; (9) Help users recognize, diagnose and recover from errors; (10) Help and documentation.

#	Frame/Page	Heuristic violated	Sev.	Description / Comment
H1	Home Dashboard	1 Visibility; 5 Error prevention	3	Budget consumption can be displayed as an absurd value (e.g. “1125005% used”) because the amount field accepts arbitrarily large input with no sanity bound or display cap.
H2	Expense Entry / Expense Detail	3 User control; 9 Error recovery	3	A recorded expense cannot be edited or deleted. A wrong amount cannot be corrected; the only “fix” is to live with the erroneous entry.
H3	Spending Alert flow	3 User control & freedom	3	Tapping an alert card opens the Notification Log instead of the goal edit form; the user cannot act on the alert directly (<i>Adjust Goal</i> not wired).
H4	New goal / Edit goal	2 Match with real world	2	Deadline field shows the raw placeholder <code>gg/mm/aaaa</code> rather than a localized, readable date hint.
H5	Home Dashboard	8 Aesthetic & minimalist	1	Alert banner and “goal at risk” card duplicate the same information.
H6	Home Dashboard	6 Recognition rather than recall	1	A newly recorded expense is not shown on the dashboard; the user must open Reports. (Not a specified requirement; recorded as a design suggestion.)

Table 10.1: Complete Heuristic Evaluation form for Prototype Zero.

10.2 Cognitive Walkthrough — Full Form

Evaluators: Jacopo Rossi, Fabiano Cacioli, Emanuele Smisi, Fabrizio Pietrobono, Luca Buonomini **Prototype:** Zero **Date:** 25 May 2026

Task. “Record a new expense of 20 € in the Transport category and save it.”

Users. Target profile: aged 18–34, students and young workers, habitual smartphone users, familiar with mobile apps but not with formal budgeting tools.

The four questions (asked for every action). (1) Is the effect of the action the same as the user’s goal at that point? (2) Will users see that the action is available? (3) Once users have found the correct action, will they know it is the one they need? (4) After the action is taken, will users understand the feedback they get?

Act. 1 — tap “+ Expense”. *Resp. 1: the New Expense modal opens at the amount step.*

- Q1: Yes — adding an expense is the user’s immediate goal.
- Q2: Yes — “+ Expense” is the largest, primary-coloured action on the dashboard.
- Q3: Yes — the label and icon make its purpose unambiguous.
- Q4: Yes — the modal opening confirms the action.

Act. 2 — type “20”, tap “Next”. *Resp. 2: the modal advances to the category list.*

- Q1: Yes — entering the amount is the natural first sub-goal.
- Q2: Yes — the field is focused on open; “Next” is visible.
- Q3: Yes — the currency prefix and the numeric field are clear.
- Q4: Yes — advancing to the category list signals progress.

Act. 3 — tap “Transport”. *Resp. 3: the category is ticked and the modal advances directly to the review screen (default currency applied).*

- Q1: Yes — choosing a category matches the user’s goal.
- Q2: Yes — the category list is clearly presented.
- Q3: Yes — the selected row is highlighted and ticked.
- Q4: **Partial (documented negative)** — tapping the category silently skips the currency step and jumps to review; a first-time user may not immediately realise the flow has advanced. The explicit “default” label on the currency row mitigates this. Recorded as a minor feedback observation.

Act. 4 — tap “Save”. *Resp. 4: a confirmation screen with an animated checkmark is shown.*

- Q1: Yes — saving is the user’s final goal.
- Q2: Yes — “Save” is the prominent primary action on the review screen.
- Q3: Yes — the review restates amount, category, and currency; “Edit” offers a recovery path.
- Q4: Yes — the animated checkmark confirms success.

Act. 5 — (no action) auto-return after 1.5 s. *Resp. 5: the app returns to the dashboard.*

- Q1: Yes — returning to the dashboard matches the user’s expectation after saving.
- Q4: Yes — the “Returning to dashboard...” copy sets the expectation for the automatic transition.

Outcome. All actions pass; the only documented negative is the partial feedback answer at Act. 3. The task is judged highly learnable.

10.3 Think-Aloud — Task List and Raw Observations

The three tasks administered in the qualitative study:

1. Record a new expense and assign it to a category.
2. Create a new savings goal with a deadline and add a first deposit to it.
3. Open the monthly report and find the single most expensive transaction.

Sessions were moderated with minimal intervention; the moderator explained that the interface (not the participant) was being tested, that the participant could stop at any time, and that the moderator could not help. Hesitations, wrong turns, spontaneous comments, and completion were recorded for each of the five participants. All five completed all three tasks.

P	All tasks done	Task 3 time	Observations
P1	Yes	57 s	No hesitation on any task.
P2	Yes	47 s	One mis-click on Task 2: tapped “Add” on an existing goal instead of the intended deposit.
P3	Yes	61 s	Hesitated on goal priority; asked “if I set High, what really changes — just the colour?”
P4	Yes	55 s	No issues.
P5	Yes	49 s	No issues; volunteered that they liked the app.

Table 10.2: Think-Aloud raw observations (five participants).

10.4 Controlled Experiment — Raw Data and Analysis

- **Hypothesis.** H_1 : deposit-button placement affects task time. H_0 : no difference.
- **Independent variable.** Button placement: A (on the goal card) vs. B (inside goal detail).
- **Dependent variables.** Task-completion time (s); number of mis-taps.
- **Controlled variables.** Task wording, seed data, device, starting screen.
- **Design.** Between-groups; twelve subjects matching the target profile, randomly split into two equal groups of six; goal state reset before each participant; same experimenter timing every run.
- **Task.** “Add a 50 € deposit to the *trip* goal.”

#	Group A — card (s)	Group B — detail (s)
1	11.50	7.77
2	6.17	10.00
3	14.00	16.50
4	10.70	11.90
5	8.10	9.90
6	5.90	12.00
Mean	9.40	11.35
Mis-taps	1	0

Table 10.3: Raw completion times for the deposit task, by group.

Analysis. One-way ANOVA at $\alpha = 0.05$ (full table in Section 6.2.3, Table 6.4): $SS_{\text{between}} = 11.41$, $SS_{\text{within}} = 95.62$, $df = (1, 10)$, giving $F = 1.19$. Since $F < F_{\text{crit}}(1, 10) = 4.96$ (equivalently $p = 0.30 > 0.05$), the null hypothesis is *not* rejected: no significant difference in completion time between the two designs. The simpler existing design (variant B) is therefore retained.